

American Board of Family Medicine



IN-TRAINING EXAMINATION

1. Your practice management team has reviewed the clinic-wide cancer screening rates and found that the frequency of cervical cancer screening appears to have fallen behind other cancer screenings. You are asked to lead a quality improvement project to increase rates of cervical cancer screening for appropriate patients.

Which one of the following would be the most appropriate first step?

- A) Establishing a system for reviewing charts and contacting patients who are overdue for cervical cancer screening
 - B) Evaluating barriers to cervical cancer screening and reporting
 - C) Placing posters with information about cervical cancer screening inside the clinic
 - D) Sending generalized reminders to patients who may qualify for cervical cancer screening
 - E) Sending generalized reminders to all clinicians regarding cervical cancer screening guidelines
2. You are seeing a 32-year-old male in the office. His most recent hemoglobin A_{1c} was 6.1%, indicating prediabetes.

Which one of the following is most likely to prevent the development of diabetes?

- A) Intensive lifestyle and behavior change program
- B) Low-carbohydrate diet
- C) Reduction in daily caloric intake by 250 kcal per day
- D) Strength training for 90 minutes per week

3. An otherwise healthy 47-year-old male presents to the urgent care clinic with pain and swelling of his right foot after sustaining an injury during a game of ultimate frisbee. He jumped to catch the frisbee and landed on uneven ground. He is unsure of how he landed, but thinks he may have twisted his right foot, causing him to trip and fall. He has significant pain and swelling of his right foot but is weight-bearing. He rates the pain as 10 on a scale of 10.

On examination, his vital signs are unremarkable. His right lateral foot shows significant bruising and swelling. The skin is intact. Range of movement is impaired due to the swelling and discomfort, and sensation and circulation are intact. An x-ray of the right foot is shown below.

Which one of the following is the most appropriate next step in management?



- A) Weight-bearing with a walking boot for 3–6 weeks
- B) Non-weight-bearing with a walking boot for 3–6 weeks
- C) Non-weight-bearing with a short leg cast for 6 weeks
- D) Referral to a physical therapist
- E) Referral to an orthopedic surgeon

4. A 78-year-old female presents to the emergency department with a 2-day history of cough and fever and a 2-hour history of confusion. Her vital signs include a weight of 67 kg (148 lb), a temperature of 38.8 °C (101.8 °F), a pulse rate of 122 beats/min, and a blood pressure of 72/44 mm Hg with a mean arterial pressure (MAP) of 53 mm Hg. A chest x-ray shows left lower lobe pneumonia. She is given 2 L of lactated Ringer solution and started on broad-spectrum antibiotics for sepsis secondary to pneumonia. Her blood pressure 3 hours later is 82/46 mm Hg with a MAP of 58 mm Hg.

Which one of the following is the most appropriate immediate next step in management?

- A) A 1-L bolus of normal saline
 - B) A dopamine infusion
 - C) A norepinephrine infusion
 - D) A vasopressin infusion
 - E) Placement of a central line
5. You are called to the emergency department (ED) to admit a 48-year-old female with a past medical history of hyperthyroidism who presented with agitation, palpitations, and diarrhea. The ED team diagnosed thyroid storm secondary to the abrupt discontinuation of the patient's methimazole 2 weeks ago due to a lapse in insurance.

On examination, she is tachycardic and hypertensive but can sustain a conversation without dyspnea. Laboratory studies are significant for elevated free T₃ and T₄ levels and an undetectable TSH level. A urine pregnancy test is negative.

Which one of the following would be the most appropriate initial step in management?

- A) Inorganic iodine
- B) Prednisone
- C) Propylthiouracil
- D) Plasmapheresis

6. According to guidelines from the American Gastroenterological Association, which one of the following is recommended as the best pharmacologic treatment of typical symptoms in diarrhea-predominant irritable bowel syndrome?
- A) Amitriptyline
 - B) Dicyclomine
 - C) Linaclotide (Linzess)
 - D) Lubiprostone (Amitiza)
 - E) Rifaximin (Xifaxan)
7. A 56-year-old female with moderately controlled type 2 diabetes presents with painful diabetic neuropathy in her feet. You address her diabetic control and other risk factors, but she says the pain continues and is affecting her life adversely.

Which one of the following is considered a first-line treatment for pain relief in this condition?

- A) Topical capsaicin
- B) Topical lidocaine
- C) Oral gabapentin
- D) Oral hydrocodone
- E) Peripheral transcutaneous electrical nerve stimulation

8. A generally healthy 60-year-old female presents to your office because she is concerned about urinary leakage. Her symptoms started a couple of years ago and have worsened. The leakage occurs when she coughs or laughs, but can also happen if she waits too long to urinate. She hydrates throughout the day. She does not take any medications and does not smoke. She has a remote surgical history of a hysterectomy after delivering her third child. She does not have any fever or abdominal pain and has not been exposed to any sexually transmitted infections. Recent laboratory studies, including a CBC, comprehensive metabolic panel, TSH level, and hemoglobin A_{1c}, were unremarkable. Today she has a blood pressure of 120/70 mm Hg and a BMI of 32 kg/m².

Which one of the following would be the most appropriate next step in the workup?

- A) No testing
 - B) Urinalysis
 - C) Renal and bladder ultrasonography
 - D) Urodynamic testing
 - E) Cystoscopy
9. A 68-year-old female with previously diagnosed immune thrombocytopenia presents to your office to establish care. She has no current bleeding. A CBC is normal except for a platelet count of 47,000/mm³ (N 130,000–400,000). You refer her to a hematologist for long-term management.

Which one of the following is the most appropriate treatment for this patient's current thrombocytopenia?

- A) No treatment
- B) Oral hydroxychloroquine sulfate (Plaquenil)
- C) Oral prednisone
- D) Intravenous methylprednisolone sodium succinate (Solu-Medrol)
- E) Intravenous rituximab (Rituxan)

10. A 45-year-old male who is living with HIV and has been on highly active antiretroviral therapy (HAART) for the past 10 years presents for routine follow-up. Which one of the following would be part of an appropriate screening strategy for long-term complications associated with HAART?
- A) Calculation of the Framingham Risk Score now, and a fasting lipid profile annually
 - B) Neurocognitive screening annually
 - C) Transient elastography annually
 - D) A DXA scan annually
 - E) Urinalysis annually, and a glomerular filtration rate twice yearly
11. A 40-year-old male with type 2 diabetes presents to the emergency department with chest pain and shortness of breath. The patient has normal vital signs. A physical examination and laboratory tests, including serial troponin levels, are normal. Outpatient records from about 6 months ago include a normal cardiac stress test that was negative for myocardial ischemia.

Which one of the following would be the most appropriate next step in evaluation?

- A) Stress echocardiography
- B) Coronary CT angiography
- C) Fractional flow reserve with CT
- D) Cardiac MRI

12. A 40-year-old male with a history of alcohol use disorder presents to your office for help abstaining from alcohol. Until recently, he has been drinking at least 6 beers per day and often more on weekend days. He has 2 prior arrests for DUI, and his wife plans to file for divorce if he does not change. Over the past 2 weeks, he has been able to wean himself off alcohol and has joined a sobriety group. He requests medication to help him maintain his sobriety.

Which one of the following medications should you recommend for this patient?

- A) Acamprosate
 - B) Disulfiram
 - C) Gabapentin (Neurontin)
 - D) Topiramate (Topamax)
13. A 15-year-old male presents to your office with an intensely pruritic rash 2 days after returning from a camping trip in the woods. On examination you observe a rash on the hands, arms, face, and genitals with vesicles and a few bullae.

Which one of the following would be the most appropriate treatment option?

- A) Topical clobetasol, applied twice daily to affected areas until symptoms resolve
 - B) Oral methylprednisolone, starting at 1 mg/kg and tapering over 5 days
 - C) Oral prednisone, starting at 1 mg/kg and tapering over 15 days
 - D) Subcutaneous triamcinolone, 80 mg once
 - E) Draining the bullae and large vesicles
14. A 50-year-old male sees you for follow-up of a pulmonary nodule. He recently presented to the emergency department (ED) after a motor vehicle accident in which he sustained a broken clavicle. A CT scan in the ED showed a 6-mm, incidental, solid pulmonary nodule and he was advised to follow up with his primary care physician. Today the patient is asymptomatic other than muscle soreness. He does not have any history of smoking. An examination does not reveal any enlarged lymph nodes.

Which one of the following would be the most appropriate next step in management?

- A) No additional follow-up
- B) A repeat CT scan in 6–12 months
- C) A repeat CT scan in 18–24 months
- D) Biopsy of the nodule to determine pathology
- E) Referral to a pulmonologist

15. A 42-year-old male presents with a 3-month history of intermittent abdominal pain and fatigue. He generally feels discomfort in the upper abdomen and feels nauseated when eating. He works as a mechanic and has arthritis, for which he takes 600–800 mg of ibuprofen. He drinks 6–8 beers nightly, but does not smoke or use other substances. You order some preliminary laboratory studies, which show the following results:

Sodium	137 mEq/L (N 135–145)
Potassium	4.9 mEq/L (N 3.8–5.0)
Chloride	101 mEq/L (N 98–107)
Creatinine	1.0 mg/dL (N 0.3–1.5)
BUN	37 mg/dL (N 10–20)
Calcium	9.1 mg/dL (N 9.0–10.5)
Hemoglobin	8.8 g/dL (N 12.0–15.0)

Which one of the following would be the most appropriate next step in management?

- A) A high-dose proton pump inhibitor
- B) Endoscopy
- C) Nasogastric lavage
- D) Blood transfusion
- E) Arterial embolization

16. You see a 6-month-old infant for a routine visit. The child has started teething and the parents are concerned about preventing cavities, as their municipality does not fluoridate the water.

In addition to providing daily oral fluoride supplementation, which one of the following should you recommend?

- A) Brushing the emerging teeth daily with a rice-sized amount of fluoridated toothpaste
 - B) Using mouthwash as an adjunct to toothbrushing until age 5
 - C) Application of fluoride varnish to teeth annually, starting at age 1
 - D) Application of fluoride varnish to teeth twice yearly, starting at age 3
17. A 75-year-old male presents to your office with his family. Although he is a long-term patient of yours, you have not seen him in more than 1 year. His family tells you that they have noticed a gradual progression of neurologic symptoms over the past 6 months, including a noticeably slower gait and a tremor of his right hand. The patient says that he has noticed worsening short-term memory loss.

On examination, the patient has a flat affect and speaks slowly. You observe a shuffling gait and a resting pill-rolling tremor of his right hand. You also note cogwheel rigidity of his right arm.

Which one of the following would be the most appropriate initial medication for the treatment of this patient's condition?

- A) Amitriptyline
- B) Carbidopa/levodopa (Sinemet)
- C) Fluoxetine (Prozac)
- D) Gabapentin (Neurontin)
- E) Propranolol hydrochloride

18. A 26-year-old female presents to your clinic because of headache and vision changes. She has also noticed an increase in the size of her hands and feet, changes to her facial features, carpal tunnel symptoms, proximal muscle weakness, fatigue, and menstrual irregularities. MRI of the brain shows a 7-mm pituitary adenoma.

Which one of the following diagnostic tests is most likely to be abnormal in this patient?

- A) A dexamethasone suppression test
 - B) A serum prolactin level
 - C) A TSH level
 - D) A serum insulin-like growth factor 1 level
19. An otherwise healthy 52-year-old female with a past surgical history of hysterectomy for the treatment of uterine fibroids presents with increasing episodes of hot flashes lasting 3–5 minutes and occurring 5–10 times per day over the past 5 months. She finds the symptoms highly disruptive and is interested in options for medical therapy.

Which one of the following would be the most effective therapy for this patient's symptoms?

- A) A transdermal estradiol patch (Climara)
- B) Transdermal testosterone gel (AndroGel)
- C) Vaginal estradiol cream (Estrace Vaginal)
- D) Soy phytoestrogen supplementation
- E) Paroxetine (Paxil)

20. A 25-year-old male is brought to the level 1 trauma center with a gunshot wound. He is awake, alert, and oriented, and has full capacity for decision-making. The patient then becomes hemodynamically unstable with a blood pressure of 80/40 mm Hg, a heart rate of 140 beats/min, and a hemoglobin level of 4.0 g/dL (N 14.0–18.0). Whole blood transfusion is indicated, but the patient refuses blood products as he is a practicing Jehovah's Witness. An advance directive on file at the hospital confirms this. He is aware that not administering blood will most likely cause his death. You disagree with the patient and would like to administer a whole blood transfusion as his condition is life-threatening.

Which one of the following would be the most appropriate next step?

- A) Presenting the case to the ethics committee
 - B) Respecting the patient's wishes by avoiding transfusion
 - C) Consulting the patient's family and administering whole blood with their approval
 - D) Administering packed RBCs
 - E) Administering whole blood
21. A 72-year-old male presents with a 45-minute history of persistent bleeding from his left nostril. He has a medical history of aortic stenosis with a mechanical aortic valve replacement, coronary artery disease, hypertension, and type 2 diabetes. His medications include hydrochlorothiazide, lisinopril (Zestril), metformin, and warfarin. A physical examination reveals a blood pressure of 170/100 mm Hg without orthostatic change and a heart rate of 90 beats/min.

Which one of the following would be the most appropriate first step in the treatment of this patient's nosebleed?

- A) Digital compression of the lower nares for 20 minutes
- B) Topical vasoconstrictors
- C) Anterior nasal packing
- D) Electrical cauterization
- E) Reversal of anticoagulation

22. An 80-year-old male presents to your office for follow-up of stable angina. He has a medical history of coronary artery disease with placement of a single drug-eluting stent, and stage 4 chronic kidney disease secondary to hypertension. His most recent echocardiogram demonstrated a normal ejection fraction. He does not want additional cardiovascular intervention and prefers you to handle all of his care in the future. The patient's current medications include the following:

Aspirin, 81 mg daily
Carvedilol (Coreg), 12.5 mg twice daily
Empagliflozin (Jardiance), 10 mg daily
Losartan (Cozaar), 50 mg daily

On examination he has a blood pressure of 140/85 mm Hg, a heart rate of 52 beats/min, and an oxygen saturation of 98% on room air. A cardiopulmonary examination is within normal limits and an EKG is unchanged from past EKGs.

Which one of the following changes to this patient's medication regimen would be most appropriate to decrease his anginal symptoms?

- A) Increasing carvedilol
 - B) Increasing empagliflozin
 - C) Adding clopidogrel (Plavix)
 - D) Adding isosorbide mononitrate
 - E) Adding ivabradine (Corlanor)
23. A 35-year-old female presents because of a 4-week history of difficulty sleeping following the death of her partner of 2 years. She has difficulty initiating and maintaining sleep and reports daytime sleepiness. Her symptoms have not improved despite attending grief counseling sessions. She requests nonpharmacologic treatment for her insomnia.

Which one of the following is a component of sleep hygiene education for cognitive behavioral therapy for insomnia?

- A) Decreasing time spent in bed
- B) Increasing daytime napping
- C) Waking up at the same time every day
- D) Placing a digital clock next to the bed

24. A 32-year-old female presents with classic symptoms of mild hypothyroidism including cold intolerance, mild fatigue, and constipation. Her TSH level is 6.0 $\mu\text{U/mL}$ (N 0.4–4.2) and her free T_4 level is within normal limits. A pregnancy test is negative.

Which one of the following factors, if present, would be an indication for prescribing levothyroxine (Synthroid)?

- A) A repeat TSH level in 4–6 weeks remains at 6.0 $\mu\text{U/mL}$
 - B) A repeat TSH level in 4–6 weeks increases to 8.0 $\mu\text{U/mL}$
 - C) An elevated free T_3 level
 - D) An elevated thyroid peroxidase antibody level
 - E) Treatment-resistant constipation
25. A 44-year-old female presents to the emergency department for evaluation of heavy vaginal bleeding. She reports soaking a pad every hour for the past 6 hours and is feeling mildly short of breath and lightheaded. The patient has a past medical history of hypertension, migraines, and breast cancer for which she underwent lumpectomy 6 months ago. The patient's only medication is hydrochlorothiazide, 25 mg. The patient's vital signs include a blood pressure of 110/60 mm Hg, a heart rate of 80 beats/min, a respiratory rate of 16/min, and an oxygen saturation of 100% on room air. A prothrombin time, partial thromboplastin time, and platelet count are all normal, and a urine pregnancy test is negative, but a hemoglobin level is 8.0 g/dL (N 12.0–16.0).

Which one of the following would be the most appropriate next step?

- A) A high-dose combined oral contraceptive
- B) Intravenous conjugated estrogens (Premarin)
- C) Intravenous tranexamic acid
- D) Depot medroxyprogesterone acetate (Depo-Provera CI)
- E) A levonorgestrel IUD (Mirena)

26. A 45-year-old female presents with a 1-year history of central facial erythema, telangiectasias, and inflammatory pustules and papules. The symptoms are worsened by alcohol and excessive sun exposure. You diagnose rosacea. She has not had any previous success with moisturizers, aggressive sun protection, or mild facial soap.

Which one of the following would be the most appropriate first step in the management of this condition?

- A) Topical brimonidine (Mirvaso)
 - B) Topical cyclosporine
 - C) Topical metronidazole
 - D) Oral doxycycline
 - E) Laser or light-based therapy
27. A 67-year-old female presents for a wellness examination. She has not been seen for medical care for several years and reports that her last Papanicolaou test was 15 years ago. She does not recall any medical problems and has no known history of abnormal cervical cytology in the past.

According to the U.S. Preventive Services Task Force, which one of the following should you recommend regarding further cervical cancer screening?

- A) No further screening due to the patient's age
- B) Cervical cytology alone now, with no further screening if the result is normal
- C) Cervical cytology alone now, then repeated in 3 years with no further screening if the results are normal
- D) Cervical cytology with HPV co-testing now, with no further screening if the results are normal
- E) Cervical cytology with HPV co-testing now, then repeated in 5 years with no further screening if the results are normal

28. A 46-year-old male comes in for an acute visit for low back pain. During triage, a Patient Health Questionnaire–2 (PHQ-2) screen is positive. You then administer the PHQ-9. You ask, "Over the last 2 weeks, how often have you been bothered by thoughts that you would be better off dead, or of hurting yourself?" He responds, "Nearly every day." His total PHQ-9 score is 19.

Which one of the following would be the most appropriate next step?

- A) Suicidal ideation assessment and evaluation of risk factors for suicide completion
 - B) Escitalopram (Lexapro), and cognitive behavioral therapy
 - C) Escitalopram plus lithium, and cognitive behavioral therapy
 - D) Intravenous ketamine
 - E) Transportation to the emergency department
29. A healthy 58-year-old patient presents for an annual health maintenance examination. They had an initial screening colonoscopy 3 years ago in which 2 polyps were removed. The polyps were found to be 6-mm and 8-mm tubular adenomas. The patient does not have a family history of colon cancer.

Which one of the following is the most appropriate ongoing colon cancer screening for this patient?

- A) No further follow-up
 - B) Fecal immunochemical testing (FIT) now
 - C) Stool DNA testing (Cologuard) now
 - D) Repeat colonoscopy within the next 2 years
 - E) Repeat colonoscopy 7–10 years after the initial colonoscopy
30. A 27-year-old gravida 1 para 0 at 38 weeks' estimated gestation based on a first-trimester ultrasound presents for routine prenatal care. Her pregnancy has been uncomplicated. You discuss the potential risks and benefits of inducing labor at 39 weeks compared to expectant management.

In this discussion, you should tell the patient that inducing labor at 39 weeks reduces

- A) the risk of low birth weight
- B) length of stay in the labor and delivery unit
- C) risk for neonatal ICU admission
- D) the rate of cesarean delivery
- E) the risk of postpartum hemorrhage

31. A 75-year-old female presents to your office with a 1-year history of gradually worsening right-sided groin pain on standing. The pain improves with rest. These symptoms have been marginally relieved with ibuprofen and acetaminophen, but her use of these medications has been limited by gastric irritation. A physical examination is most notable for pain in the groin with passive flexion, adduction, and internal rotation (FADIR) testing.

Which one of the following treatments would be most helpful in relieving her symptoms?

- A) A topical lidocaine patch (Lidoderm)
 - B) Hydrocodone
 - C) Tramadol
 - D) Intra-articular corticosteroid injection
 - E) Intra-articular hyaluronic acid injection
32. A 65-year-old male is brought to the emergency department by his wife after developing confusion immediately after sexual intercourse. The wife reports that the patient does not remember anything from a few hours prior to the onset of symptoms. He repeatedly asks where he is, how he got here, and what time it is, despite having been given the answers to his questions. He is alert and knows his wife, his name, and his birth date. A neurologic examination is otherwise normal. A CBC and comprehensive metabolic panel are normal, an EKG is normal, and a CT scan of the head is unremarkable. It has now been 2 hours since the onset of his symptoms.

Which one of the following would be most appropriate for treatment?

- A) Observation and reassurance
- B) Oral apixaban (Eliquis)
- C) Oral aspirin and clopidogrel (Plavix)
- D) Intravenous lorazepam (Ativan)
- E) Intravenous thrombolytic therapy

33. A 55-year-old male presents to the urgent care clinic with a 24-hour history of worsening fever, chills, and body aches. He tells you that he discovered a tick on his scalp during a camping trip last week. He does not have a rash, headache, or palpitations, and a neurologic examination is negative. A Lyme enzyme immunoassay and confirmatory Western blot are positive. When he has taken doxycycline in the past, he has developed headaches, severe nausea, and vomiting.

Which one of the following would be the most appropriate treatment option for this patient?

- A) Amoxicillin
 - B) Ceftriaxone
 - C) Chloramphenicol
 - D) Doxycycline plus promethazine
34. A 66-year-old female asks about options for shingles vaccine. She received the live zoster vaccine (Zostavax) previously. She underwent urgent cholecystectomy for acute cholecystitis 2 months ago and developed shingles in a dermatomal distribution soon after hospital discharge. She was given valacyclovir (Valtrex) and gabapentin (Neurontin) for treatment and symptom relief. The rash is gone but she still has mild postherpetic neuralgia. She would like to receive the recombinant zoster vaccine (Shingrix) as soon as it is safe to receive it.

Which one of the following would be the earliest this patient can safely receive the first dose of recombinant zoster vaccine?

- A) Today
- B) After resolution of the postherpetic neuralgia
- C) 12 months after her episode of shingles
- D) Never

35. A 40-year-old male presents to the office with a 2-day history of right eye pain associated with redness and swelling. He does not wear contact lenses. Pertinent findings on a physical examination include a temperature of 37.3 °C (99.1 °F), visual acuity of 20/70 in the right eye and 20/20 in the left eye, erythema and edema of the right eyelid, and pain with extraocular movements.

Which one of the following would be the most appropriate next step?

- A) Frequent warm, moist compresses, and follow-up tomorrow
 - B) Empiric treatment with an oral antibiotic
 - C) A same-day evaluation with an ophthalmologist
 - D) Noncontrast CT of the orbits and sinuses
 - E) Transfer to the emergency department
36. A 63-year-old male presents with a 2-day history of right lower extremity edema and pain. He has a past medical history of hypertension, stage 4 chronic kidney disease, and type 2 diabetes, for which he takes insulin. His last glomerular filtration rate was 22 mL/min/1.73 m². He does not report any trauma, chest pain, or shortness of breath.

On examination he has a weight of 70 kg (154 lb) and the remainder of his vital signs are within normal limits. Examination of his right lower extremity reveals erythema, tenderness, and 2+ edema at the back of the knee. A Doppler venous duplex ultrasound of the right thigh shows a noncompressible right femoral vein.

Which one of the following would be the most appropriate initial treatment for this patient's condition?

- A) Oral dabigatran (Pradaxa), 75 mg twice daily
- B) Oral rivaroxaban (Xarelto), 15 mg twice daily for 14 days, then 20 mg daily
- C) Oral warfarin, 5 mg daily
- D) Subcutaneous enoxaparin (Lovenox), 70 mg daily
- E) Admission for intravenous heparin

37. A 65-year-old male presents to your office with a 4-month history of ill-defined lower pelvic pain. He is afebrile. An abdominal examination is unremarkable. A digital rectal examination yields a boggy prostate with moderate tenderness. Urinalysis is positive for leukocyte esterase and 10–20 WBCs/hpf, and testing for common sexually transmitted infections is negative.

Which one of the following would be the most appropriate next step in management?

- A) Amoxicillin
 - B) Ceftriaxone
 - C) Ciprofloxacin
 - D) CT of the abdomen and pelvis
 - E) Referral to a urologist
38. A 25-year-old male presents to the emergency department with a sudden onset of right-sided chest pain that started about 2 hours ago while he was resting and playing video games. He reports feeling slightly short of breath. He has no history of lung disease, and he smokes 1 pack of cigarettes per day. On examination he has a blood pressure of 120/80 mm Hg, a heart rate of 115 beats/min, a respiratory rate of 20/min, and an oxygen saturation of 90% on room air. A chest x-ray reveals a pneumothorax involving <10% of the lung space.

Which one of the following would be the most appropriate next step in management?

- A) Reassurance and discharge home with return precautions for worsening symptoms
- B) Initiation of oxygen via nasal cannula, and a repeat chest x-ray in 6 hours
- C) CT of the chest to verify x-ray findings
- D) Emergent needle decompression at the right second intercostal space in the anterior midclavicular line
- E) Placement of a chest tube at the right fourth or fifth intercostal space in the midaxillary to anterior axillary line

39. A 17-year-old female with well-controlled diabetes mellitus is brought to your office by her mother for a sports preparticipation physical evaluation prior to playing soccer. She has intermittent knee pain when running but no other symptoms.

Which one of the following, if present in this patient, would be a CONTRAINDICATION to playing soccer?

- A) A BMI $>30 \text{ kg/m}^2$
 - B) An enlarged spleen
 - C) Insulin-dependent diabetes
 - D) Mitral valve prolapse
 - E) Tibial tubercle apophysitis
40. An otherwise healthy 26-year-old female presents to your clinic with a history of 3 episodes of vertigo occurring over the last week. Each episode lasts for several hours. During the episodes, she notes fullness in the right ear with tinnitus and hearing loss. She has not noticed anything specific that triggers the episodes, and reports consuming 1 glass of wine per night. She does not report headache, visual changes, fever, or upper respiratory symptoms. She is currently asymptomatic and a physical examination is unremarkable.

Which one of the following would be the most appropriate next step?

- A) Recommending a reduction in alcohol intake
 - B) Performing the Epley maneuver
 - C) Ordering audiometry
 - D) Initiating oral hydrochlorothiazide, 25 mg daily
 - E) Initiating oral prednisone, 20 mg daily for 5 days
41. A new lipid-lowering drug is being evaluated for its cardioprotective effects. The event rate for acute myocardial infarction is 0.085 in the low-intensity treatment group and 0.045 in the high-intensity treatment group, yielding an absolute risk reduction (ARR) of 0.04.

Which one of the following is the number needed to treat (NNT)?

- A) <1
- B) 25
- C) 40
- D) 50

42. A 75-year-old female presents to your office with a 3-month history of episodes of lightheadedness on standing. Orthostatic blood pressure testing in your office reveals an initial blood pressure of 134/76 mm Hg that drops to 104/46 mm Hg on standing. A laboratory evaluation is unremarkable and review of her medications does not show an obvious cause for her symptoms. Echocardiography and tilt table testing are unremarkable. You diagnose orthostatic hypotension.

Monotherapy with which one of the following would be indicated for the management of this patient's neurogenic orthostatic hypotension?

- A) Atomoxetine (Strattera)
 - B) Fludrocortisone
 - C) Methylphenidate (Ritalin)
 - D) Midodrine
 - E) Sodium bicarbonate
43. A 22-year-old female with a past medical history of type 1 diabetes presents to the emergency department with multiple bouts of vomiting. She says that she is not consistently adherent to her insulin regimen. She has a plasma glucose level of 350 mg/dL, elevated ketones, and high anion gap metabolic acidosis. Treatment for diabetic ketoacidosis is initiated with intravenous fluids and insulin therapy.

Glucose should be added to this patient's saline infusion when her plasma glucose levels reach which one of the following ranges?

- A) <150 mg/dL
- B) 150–200 mg/dL
- C) 201–250 mg/dL
- D) 251–300 mg/dL
- E) 301–350 mg/dL

44. A 32-year-old female with a recent diagnosis of posttraumatic stress disorder presents to your clinic with difficulty sleeping. She recently left an abusive relationship and has been attending trauma-focused psychotherapy. She has found the psychotherapy helpful, but she continues to experience midsleep awakening due to hyperarousal and nightmares.

Which one of the following would be the most appropriate treatment?

- A) Amitriptyline
 - B) Mirtazapine (Remeron)
 - C) Prazosin
 - D) Risperidone (Risperdal)
 - E) Temazepam (Restoril)
45. A 58-year-old female with type 2 diabetes controlled on metformin recently underwent bariatric surgery. Today, she reports feeling off balance and having a burning sensation in her bilateral hands and feet that is worse at night. She reports difficulty ambulating and has fallen several times over the past few weeks. She has intermittent nausea, vomiting, and confusion, and reports aches and pains all over. A physical examination is significant for absent patellar and Achilles reflexes. Her strength is 4/5 in the bilateral lower extremities with decreased pinprick sensation.

Which one of the following is the most likely diagnosis?

- A) Diabetic peripheral neuropathy
- B) Lyme disease
- C) Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS)
- D) Secondary syphilis
- E) Vitamin B₁ deficiency (beriberi)

46. A 76-year-old female with a past medical history of hypertension and major depressive disorder presents because of acute right knee pain that began after she sustained a mechanical fall and landed on her knee while taking a walk near her house yesterday. She was able to walk home without difficulty. She applied ice to the area after the fall, but her knee was more painful this morning. When asked to point to the pain, she points to an area lateral to the inferior patella.

Examination of the knee reveals no redness or swelling but minor warmth over the anterior knee. Lower extremity sensation is intact. The patient has full range of motion with pain. There is no tenderness of the patella or the fibular head with palpation. Provocative testing including an Apley grind test, a McMurray test, a Lachman test, and anterior and posterior drawer tests is negative.

Which one of the following would be the most appropriate next step to aid in diagnosis?

- A) No further testing or imaging
 - B) Ultrasonography
 - C) Radiography
 - D) MRI
 - E) Arthrocentesis with culture and Gram stain
47. A 31-year-old gravida 3 para 2 at 8 weeks' gestation presents to the urgent care clinic because of a 2-day history of headache and fever. This morning she removed an engorged tick from her leg. On examination she is ill appearing but alert and interactive. An HEENT examination is normal and there is no pain with neck flexion. A cardiopulmonary examination is normal. There is no rash. You suspect Rocky Mountain spotted fever.

In addition to supportive care, which one of the following would be the most appropriate treatment?

- A) No additional treatment
- B) Amoxicillin
- C) Chloramphenicol
- D) Doxycycline
- E) Erythromycin

48. A 53-year-old female with no significant past medical history presents to the emergency department with fatigue, pruritus, and right upper quadrant discomfort. A physical examination reveals mild hepatomegaly.

Laboratory Findings

Alkaline phosphatase	450 U/L (N 44–147)
AST	70 U/L (N 10–40)
Total bilirubin	1.5 mg/dL (N 0.1–1.2)
Antimitochondrial antibodies	positive

Which one of the following is the most likely diagnosis?

- A) Autoimmune hepatitis
- B) Drug-induced liver injury
- C) Metabolic dysfunction–associated steatotic liver disease (MASLD) (nonalcoholic fatty liver disease)
- D) Primary biliary cholangitis
- E) Primary sclerosing cholangitis

49. A generally healthy 28-year-old female presents to your office because she is concerned about changes in her mood. For the last few months she has been intermittently irritable and unable to sit still. She has had some weight loss, is not sleeping well, and feels fatigued. She has been working extra hours to get ahead of her office deadlines. She has maxed out her credit card and repeatedly cheated on her husband, who filed for divorce 2 weeks ago. She now feels depressed and anxious, and has considered suicide but does not have suicidal ideations today. She says that she has had similar episodes in the past. Her father left when she was an infant and her mother is incarcerated, and the patient is unsure of their medical history.

Today the patient's vital signs, a CBC, a comprehensive metabolic panel, and TSH and vitamin B₁₂ levels are unremarkable. A urine pregnancy test and drug screen are negative. She is alert and oriented during the office visit.

Which one of the following would be the most appropriate therapy at this time?

- A) Bupropion
- B) Paroxetine (Paxil)
- C) Quetiapine (Seroquel)
- D) Sertraline (Zoloft)
- E) Electroconvulsive therapy

50. A 23-year-old female presents for IUD removal because she wants to conceive. She has never been pregnant, is in good health, and takes no medications. Her blood pressure is normal and her BMI is 33 kg/m².

Which one of the following should you recommend regarding aspirin prophylaxis for preeclampsia?

- A) No aspirin as she does not meet criteria
- B) Aspirin for 1–3 months prior to conception
- C) Aspirin at 12 weeks' gestation
- D) Aspirin if her blood pressure during pregnancy is >135/80 mm Hg

51. A 74-year-old female presents to your office for follow-up of cardioembolic stroke 1 day after being discharged from the hospital. She presented to the hospital with acute left-sided hand and face weakness. An evaluation in the hospital revealed new atrial fibrillation, mild mitral regurgitation, and a small right-sided cerebral infarction. Cardioembolic stroke was diagnosed. She was discharged 48 hours later after her left-sided weakness had almost completely resolved.

Which one of the following therapeutic options is preferred for this patient?

- A) Apixaban (Eliquis)
 - B) Aspirin
 - C) Clopidogrel (Plavix)
 - D) Warfarin
 - E) Aspirin plus clopidogrel
52. A 75-year-old female with an ST-elevation myocardial infarction (STEMI) diagnosed in the emergency department has been admitted to your service. She underwent a percutaneous coronary intervention with a drug-eluting stent to the circumflex artery and had been recovering well until she developed acute shortness of breath on the fourth day of hospitalization. Today, her vital signs reveal a blood pressure of 91/55 mm Hg, a heart rate of 110 beats/min, and an oxygen saturation of 89% on room air. On physical examination, the patient appears moderately distressed and has no jugular venous pulse. Auscultation reveals a regular rate, normal S₁ and S₂, and a systolic murmur over the mitral post. Crackles are heard in all lung fields. There is no lower extremity edema.

Which one of the following is the likely etiology of this condition?

- A) Cardiac tamponade
- B) Papillary muscle rupture
- C) Pericarditis
- D) Pulmonary embolism
- E) Supraventricular tachyarrhythmia

53. A 14-year-old male sees you to establish care. He has a past medical history of asthma and uses an albuterol inhaler (Proventil, Ventolin) as needed. He normally has symptoms about 3–4 days per week. Recent pulmonary function tests showed mildly reduced lung function.

Based on the Global Initiative for Asthma (GINA) guidelines, which one of the following would be the preferred treatment strategy for this patient?

- A) Continuing as-needed albuterol
 - B) Switching to as-needed formoterol
 - C) Switching to an as-needed low-dose inhaled corticosteroid (ICS) plus formoterol
 - D) Continuing as-needed albuterol, and adding a low-dose ICS for maintenance
 - E) Continuing as-needed albuterol, and adding a low-dose ICS plus salmeterol for maintenance
54. A 37-year-old male who recently returned from a trip to Vietnam presents for evaluation of an intensely pruritic rash on his feet. Physical examination findings are normal other than an erythematous, serpiginous rash on his feet and lower legs.

Which one of the following is the most likely diagnosis?

- A) Cutaneous larva migrans
 - B) Cutaneous leishmaniasis
 - C) Herpesvirus B
 - D) *Neisseria meningitidis*
 - E) Scabies
55. A 10-year-old male is brought to your office by his mother for follow-up of treatment for a *Cryptosporidium* infection that he likely contracted during a camping trip after swimming in a freshwater lake. He has completed treatment and is asymptomatic. His mother asks when he is allowed to resume swimming.

Because of concerns for ongoing asymptomatic shedding, you should recommend that the patient may return to swimming

- A) 3 days after treatment is completed
- B) 1 week after treatment is completed
- C) 3 days after symptoms resolve
- D) 1 week after symptoms resolve
- E) 2 weeks after symptoms resolve

56. A previously healthy 27-year-old female at 36 weeks' gestation presents to the emergency department (ED) in a rural town. She reports a 1-week history of headaches, nausea, fatigue, and right upper quadrant pain. She has been taking acetaminophen for pain with minimal improvement in her symptoms, which have worsened today. On examination she has a temperature of 36.8 °C (98.2 °F), a blood pressure of 150/95 mm Hg, a respiratory rate of 18/min, and a pulse rate of 104 beats/min. There are no signs of distress on neonatal tracing.

Laboratory Findings

WBCs	12,400/mm ³ (N 4000–11,000)
Hemoglobin	9.8 g/dL (N 12.0–15.0)
Platelets	43,000/mm ³ (N 150,000–400,000)
AST	112 U/L (N 10–59)
ALT	95 U/L (N 10–28)
Serum creatinine	0.8 mg/dL (N 0.6–1.1)
Urine protein to creatinine ratio	422 mg/g (N <300)

Which one of the following would be the most appropriate next step?

- A) Administering oral prednisone
 - B) Administering *N*-acetylcysteine
 - C) Consultation with a gastroenterologist
 - D) Arranging for urgent delivery
57. A 64-year-old female with a history of hypertension and no prior evidence of kidney disease presents for follow-up of a recent basic metabolic panel that showed an estimated glomerular filtration rate (eGFR) of 49 mL/min/1.73 m². Which one of the following would confirm a diagnosis of chronic kidney disease?
- A) A urine albumin to creatinine ratio >30 mg/g at this time
 - B) A urine albumin to creatinine ratio >300 mg/g at this time
 - C) An additional eGFR <60 mL/min/1.73 m² in 1 month
 - D) An additional eGFR <90 mL/min/1.73 m² in 2 months
 - E) An additional eGFR <60 mL/min/1.73 m² in 3 months

58. A 4-year-old male presents with a 6-day history of fever, rash, and an enlarged lymph node. On examination, you note erythema of the lips, bilateral conjunctivitis, and a diffuse maculopapular rash on the trunk and extremities. The right anterior cervical lymph node is nontender and 2 cm in size.

Which one of the following is the most serious complication of this patient's condition?

- A) Cardiac
- B) Hematologic
- C) Hepatic
- D) Pulmonary
- E) Renal

59. A 40-year-old female presents for a health maintenance examination. Her medical history is unremarkable, but her family history is significant for colorectal cancer in her father, thyroid cancer in her paternal aunt, and endometrial cancer in her maternal grandmother. The patient says that her friend was diagnosed with breast cancer 2 weeks ago and is *BRCA2* positive. She is anxious about her risk of breast cancer and requests a mammogram and *BRCA* testing.

According to the U.S. Preventive Services Task Force guidelines, which one of the following would be most appropriate for this patient regarding genetic testing for *BRCA1/2*?

- A) Reassurance only
- B) Risk assessment using the 7-Question Family History Screening Tool
- C) Testing for *BRCA1/2* mutation
- D) Genetic counseling

60. You are seeing a patient at 12 weeks' gestation for prenatal care. She has a family history of neural tube defects and is taking 4 mg of high-dose folic acid supplementation daily. Her pregnancy is progressing normally, and she has no concerns.

Which one of the following should you recommend for this patient regarding ongoing folic acid supplementation?

- A) Continuing with 4 mg daily for the remainder of the pregnancy
 - B) Decreasing the dosage to 1 mg daily for the remainder of the pregnancy
 - C) Decreasing the dosage to 0.4 mg daily for the remainder of the pregnancy
 - D) Discontinuing folic acid
61. A 55-year-old male with a history of diabetes mellitus, hypertension, and stage 3 chronic kidney disease presents for management of recurrent gout. He has had multiple attacks this year despite dietary changes. His uric acid level is 8.6 mg/dL (N 4.5–8.0) and he would like to start gout prophylaxis.

Which one of the following would be most appropriate to initiate at this time?

- A) Allopurinol (Zyloprim)
 - B) Colchicine (Colcrys)
 - C) Febuxostat (Uloric)
 - D) Pegloticase (Krystexxa)
 - E) Allopurinol plus colchicine
62. In a patient with aortic stenosis, which one of the following is an indication for referral for possible valve replacement?
- A) Calcification of the valve on an echocardiogram
 - B) Development of mitral insufficiency on an echocardiogram
 - C) Left ventricular hypertrophy noted on a chest x-ray
 - D) A systolic ejection murmur that has increased from grade 1/6 to grade 2/6
 - E) New-onset angina

63. A 23-year-old male presents on a spring day with a 2–3-day history of mild nasal congestion, sneezing, and slight watery discharge. He does not have a fever or other significant concerns.

Which one of the following would be the most appropriate treatment?

- A) Intranasal antihistamines
- B) Intranasal corticosteroids
- C) Intranasal cromolyn
- D) Oral antibiotics
- E) Oral corticosteroids

64. A 30-year-old female presents for follow-up of infertility. She and her husband have been trying to conceive for the past 12 months. She reports a history of irregular menstrual bleeding over the past several years. A physical examination is unremarkable except for a BMI of 29 kg/m². You suspect polycystic ovary syndrome (PCOS) and recommend diagnostic testing with fasting laboratory studies scheduled for 8:00 a.m.

An elevation in which one of the following hormones would most strongly support a diagnosis of PCOS in this patient?

- A) 17-Hydroxyprogesterone
- B) Cortisol
- C) Glucose
- D) Insulin
- E) Testosterone

65. A generally healthy 36-year-old male presents to your clinic with a 36-hour history of dizziness and nausea. He says he feels as though the world is spinning and he tends to fall to the left when he tries to walk. The episodes of dizziness occur spontaneously and can last from minutes to hours. He does not have any recent illness, hearing loss, or tinnitus, and he does not take any medications.

An HEENT examination is normal. On neurologic testing, he exhibits horizontal nystagmus with left lateral gaze. Head-impulse testing reveals a rapid movement of both eyes.

Which one of the following would be the most appropriate next step?

- A) Home Epley maneuvers
 - B) Reassurance and antiemetics
 - C) Prednisone, 40 mg daily for 5 days
 - D) Valacyclovir (Valtrex), 1000 mg twice daily for 7 days
 - E) MRI of the brain
66. A 24-year-old gravida 2 para 2 presents to your office with melasma that has not resolved since she gave birth 6 months ago. She is eager to be treated.

Which one of the following topical treatments would be the most effective for this patient's melasma?

- A) Azelaic acid
- B) Hydroquinone
- C) Imiquimod (Aldara, Zyclara)
- D) Benzoyl peroxide plus a corticosteroid
- E) A combination of fluocinolone, hydroquinone, and tretinoin

67. You see a newborn for a routine wellness check. The parents tell you that there is a family history of allergies to eggs and peanuts, and they ask how to reduce their child's risk of developing these food allergies.

Evidence supports exposing the patient to these potentially allergenic foods starting at which one of the following ages?

- A) 4–6 months
 - B) 9–12 months
 - C) 1–2 years
 - D) 3–5 years
 - E) >5 years
68. A 23-year-old female presents after a sexual assault that occurred the prior evening. She reports nonconsensual penetrative vaginal intercourse. The assailant was an acquaintance with unknown medical history. The patient takes no medications and is up to date on vaccinations, including tetanus, HPV, and hepatitis B, although her hepatitis B serology status is unknown. She elects to undergo a medical forensic examination, which is completed with the help of a certified Sexual Assault Nurse Examiner. A urine pregnancy test is negative, and sexually transmitted infection testing is performed.

Which one of the following would be most appropriate to offer at this time?

- A) Emtricitabine/tenofovir alafenamide (Descovy)
- B) Norgestimate/ethinyl estradiol (Sprintec)
- C) Sofosbuvir/velpatasvir (Epclusa)
- D) Azithromycin (Zithromax), ceftriaxone, and metronidazole (Flagyl)
- E) The etonogestrel implant (Nexplanon)

69. A 29-year-old female presents with a long-standing history of episodes of depressed mood, headaches, and irritability that occur in the 1–2 weeks preceding her menses. She has a history of anxiety that is well controlled on fluoxetine (Prozac), 60 mg daily. She has a copper IUD (Paragard) for contraception. She has tried combined oral contraceptives to control the premenstrual symptoms, but found that they worsened her irritability.

Which one of the following treatments would be most effective for this patient's symptoms?

- A) Extended-release bupropion, 150 mg daily
 - B) Calcium carbonate, 600 mg twice daily
 - C) Dutasteride (Avodart), 0.5 mg daily
 - D) Vitamin D₃ (cholecalciferol), 1000 IU daily
 - E) Acupuncture
70. A 72-year-old female with osteoporosis (T-score –2.8) is seen in your office for the first time. She was prescribed alendronate (Fosamax), 70 mg weekly, by her previous physician. The patient took the medication but discontinued it after 4 weeks because of gastrointestinal symptoms.

Which one of the following would be the most appropriate action at this time?

- A) Discontinuing the medication only
 - B) Instructing the patient to take the medication before bed
 - C) Instructing the patient to take the medication with food
 - D) Discontinuing the medication and starting raloxifene (Evista)
 - E) Discontinuing the medication and arranging for intravenous bisphosphonates
71. A 27-year-old patient presents to the emergency department with a fever, headache, and stiff neck. You diagnose sepsis and perform a lumbar puncture. The results are consistent with bacterial meningitis.

In addition to antibiotics, which one of the following intravenous therapies is most likely to improve mortality?

- A) Acyclovir
- B) Dexamethasone
- C) Furosemide
- D) Glycerol
- E) Immunoglobulin

72. A 33-year-old male presents for follow-up of long-standing heartburn. His symptoms have been present for 3 years and recent treatment for 1 month with pantoprazole (Protonix) has only provided partial relief. He has a history of allergic rhinitis that is treated with fluticasone propionate nasal spray (Flonase). He has had dysphagia with solids, and a food impaction 3 months ago that resolved spontaneously. Upper endoscopy reveals eosinophilic esophagitis.

Which one of the following is most likely to improve this patient's symptoms?

- A) A clotrimazole oropharyngeal lozenge
 - B) Budesonide oral suspension (Eohilia)
 - C) Oral famotidine (Pepcid)
 - D) Oral metoclopramide (Reglan)
 - E) Oral montelukast (Singulair)
73. A 45-year-old male with a recent diagnosis of Crohn disease presents to your office for a health maintenance examination. His gastroenterologist advised that he ask you about vaccine updates. He has not seen a primary care physician in over 15 years. His current medications include methotrexate and azathioprine (Imuran). He received a Tdap vaccine 6 years ago at an urgent care clinic and recently received a seasonal influenza vaccine and an updated COVID-19 vaccine at his gastroenterologist's office. He is confident that he received all of his childhood vaccines.

Which one of the following vaccines should you recommend today?

- A) 20-valent pneumococcal conjugate vaccine (PCV20, Prevnar 20)
- B) Meningococcal
- C) MMR
- D) Tdap
- E) Zoster vaccine (Shingrix)

74. A 19-year-old college student with a recent history of sinusitis and a past medical history of depression, adrenal insufficiency, and hypothyroidism is brought to the urgent care clinic by her roommates after she experienced a syncopal episode. The patient reports progressive weakness and fatigue, increasingly severe abdominal pain, and repeated episodes of nausea and vomiting over the last few days. A review of her electronic health record shows the following medications:

Amoxicillin/clavulanate (Augmentin)
Escitalopram (Lexapro)
Fludrocortisone
Levothyroxine (Synthroid)
Prednisolone

Her vital signs at presentation include a blood pressure of 97/58 mm Hg, a pulse rate of 116 beats/min, and a temperature of 37.2 °C (99.0 °F). A basic metabolic panel is notable for the following results:

Sodium	133 mEq/L (N 136–145)
Potassium	5.8 mEq/L (N 3.4–4.4)
Creatinine	1.0 mg/dL (N 0.6–1.1)
Glucose	60 mg/dL

Which one of the following therapies should be started as soon as possible to prevent further worsening of this patient's acute condition?

- A) Intramuscular glucagon, 1 mg
- B) Intravenous dexamethasone, 1 mg
- C) Intravenous hydrocortisone (Solu-Cortef), 100 mg
- D) Intravenous levofloxacin, 500 mg
- E) Intravenous ondansetron, 4 mg

75. A 37-year-old female is brought to the emergency department by ambulance after collapsing while standing in line at a bookstore. Her friend witnessed the episode and describes it as "fainting." A history and physical examination were performed at the bedside while triaging the patient. The patient's past medical history and a physical examination, including vital signs and a neurologic assessment, are unremarkable.

Which one of the following would be the most appropriate next step in the evaluation of this patient with suspected syncope?

- A) No further testing
 - B) An EKG
 - C) An EEG
 - D) Carotid ultrasonography
 - E) Head CT
76. A 64-year-old male presents with urinary frequency. His symptoms have been ongoing for months, but he says they have acutely worsened in the last 6–8 weeks. His symptoms are predominantly storage-related with daytime frequency, nocturia, urgency, and even infrequent incontinence. He also reports difficulty voiding, noting a weak stream and hesitancy. He does not have erectile dysfunction. Urinalysis is negative.

In addition to behavioral and lifestyle alterations, which one of the following medications should you recommend for initial therapy?

- A) A 5- α -reductase inhibitor
- B) An α -antagonist
- C) An anticholinergic
- D) A β_3 -agonist
- E) A phosphodiesterase-5 inhibitor

77. A 5-day-old male is brought to your office by his mother for an initial well child visit. After an uncomplicated, full-term pregnancy, the mother gave birth 102 hours ago through vaginal delivery and was discharged 48 hours ago. She shares that this is her first child, and she is concerned that he is not getting enough milk despite exclusively breastfeeding 10–12 times per day.

Which one of the following factors, if present, would indicate inadequate breastfeeding at this time?

- A) Four green stools per day
 - B) Four wet diapers per day
 - C) Infant weight loss of 5% from birth weight
 - D) Maternal breasts that appear small and not full
 - E) Maternal nipple pain during the first 15 seconds of each feed
78. A 25-year-old male presents to the urgent care clinic with a 4-day history of fever, chills, myalgias, and cough. He lives with a roommate and works remotely from home. His last influenza vaccine was 5 years ago, and his most recent COVID-19 vaccine was 2 years ago. His temperature is 38.3 °C (100.9 °F), and his heart rate and blood pressure are within the normal range. A physical examination is normal except for occasional rhonchi. Rapid antigen testing is positive for influenza B and negative for influenza A and COVID-19.

Which one of the following is the most appropriate treatment for this patient's influenza B?

- A) Supportive therapy
 - B) Methylprednisolone (Medrol)
 - C) Oseltamivir (Tamiflu)
 - D) Rimantadine
79. A 30-year-old male with a long-standing history of obsessive-compulsive disorder presents for follow-up to discuss potential pharmacotherapy due to an inadequate treatment response to psychotherapy. Which one of the following medications would be most likely to improve this patient's symptoms while minimizing the risk of adverse side effects?
- A) Bupropion
 - B) Buspirone
 - C) Clomipramine (Anafranil)
 - D) Quetiapine (Seroquel)
 - E) Sertraline (Zoloft)

80. A 76-year-old female with hypertension sees you for a routine follow-up visit. She fell about 2 weeks ago while hurrying to answer her home telephone, and sustained bruising and swelling of the left lower leg. She does not recall feeling dizzy or lightheaded before the fall, and she did not hit her head or lose consciousness. She was able to get up and walk afterward without any limitations but developed a large bruise over the lateral hip. She has not had any previous falls and she feels well today. She lives alone and performs activities of daily living independently.

Which one of the following has the best evidence to prevent future falls and associated serious injury?

- A) Patient education on fall harms and prevention
 - B) Use of a cane while walking
 - C) Referral to a group exercise program, including gait, balance, and functional training
 - D) Referral to an ophthalmologist for a comprehensive vision assessment
 - E) Referral to a podiatrist for footwear assessment
81. Which one of the following would be most appropriate for initial treatment of *Helicobacter pylori* infection?
- A) Dual therapy with vonoprazan (Voquezna) and amoxicillin
 - B) Triple therapy with omeprazole, amoxicillin, and clarithromycin
 - C) Triple therapy with omeprazole, amoxicillin, and levofloxacin
 - D) Quadruple therapy with bismuth, omeprazole, metronidazole, and tetracycline
82. The parent of a 5-year-old male calls your office after the child sustains burns to his arms and chest while helping his father boil pasta in the kitchen. The father tells you that the burned areas of skin are red without any open areas. He estimates the size of the burns on the arms at 2–3 cm in diameter and those on the chest at 5–6 cm. The child is crying in pain and the father asks what he should do.

Which one of the following immediate interventions has the best evidence of reducing burn depth, improving healing time, and decreasing skin graft requirements?

- A) Cooling the burn surfaces with running tap water
- B) Applying ice to the burn surfaces
- C) Covering the burn surfaces with cool, wet gauze
- D) Applying aloe vera gel to the burn surfaces and leaving them uncovered
- E) Applying diclofenac gel to the burn surfaces and leaving them uncovered

83. A 45-year-old female with a past medical history of obesity and cigarette smoking presents with a popliteal deep vein thrombosis of the right leg, confirmed on ultrasonography. She has no prior history of venous thromboembolism and has not had any recent surgery, trauma, or immobilization. She does not take any medications.

Which one of the following anticoagulation regimens would be the most appropriate initial treatment?

- A) A direct-acting oral anticoagulant (DOAC) only
 - B) Low-molecular-weight heparin (LMWH) only
 - C) Warfarin only
 - D) LMWH for 7 days, followed by warfarin
 - E) Unfractionated heparin for 48 hours, followed by a DOAC
84. Which one of the following puts a patient at the highest risk of developing breast cancer?
- A) A first pregnancy at <30 years of age
 - B) Onset of menopause at <50 years of age
 - C) Fatty breasts on mammography
 - D) A grandmother with a history of breast cancer
 - E) Chest radiation therapy as a child
85. An 83-year-old female sees you for follow-up 2 weeks after being discharged home from an acute rehabilitation facility. She experienced an acute ischemic stroke 6 weeks ago and spent several weeks in the rehabilitation setting after an initial hospital stay. Home health services have been assisting her since she returned home. Today she reports improved mobility, speech, and swallowing, but persistent difficulties with low mood. She finds it difficult to enjoy hobbies and avoids visiting with family and friends. She has been sleeping more than usual and finds it difficult to get out of bed in the morning. You diagnose poststroke depression.

In addition to talk therapy, which one of the following is recommended as first-line therapy for this condition?

- A) Bupropion
- B) Escitalopram (Lexapro)
- C) Nortriptyline (Pamelor)
- D) Repetitive transcranial magnetic stimulation

86. Patients with which one of the following should be provided opioid overdose education and a prescription for naloxone?

- A) A history of substance use disorder
- B) A prescription for alprazolam (Xanax), 1 mg nightly as needed
- C) A 3-day prescription for hydrocodone/acetaminophen, 5/325 mg twice daily
- D) A prescription through hospice treatment for morphine, 10 mg as needed

87. A 55-year-old female with a medical history of hypothyroidism, hypertension, and morbid obesity with a recent gastric bypass surgery presents with a history of dizziness, sweating, shaking, fatigue, and occasional fainting, occurring 1–2 hours after consuming beverages containing dairy products or a high sugar content. Her blood glucose levels, measured with her husband's glucometer during each episode, range between 50 mg/dL and 55 mg/dL.

Which one of the following is the most likely cause of this patient's symptoms?

- A) Gastroparesis
- B) Idiopathic postprandial syndrome
- C) Insulinoma
- D) Late dumping syndrome

88. A 10-month-old male is brought to your office with a 1-day history of fussiness and tugging at his ear. The child's mother says that he has had a fever this morning. He is up to date on all immunizations, and he does not have a history of previous ear infections or recent illness. He has a history of hives after being given amoxicillin.

On examination the child has a temperature of 39.2 °C (102.6 °F) but seems to be only slightly uncomfortable. When you examine his ears you note moderate bulging of the left tympanic membrane and normal findings in the right ear. All other findings are normal.

According to the American Academy of Pediatrics guidelines, which one of the following would be the most appropriate initial management?

- A) Amoxicillin
- B) Amoxicillin/clavulanate (Augmentin)
- C) Cefdinir
- D) Ciprofloxacin (Cipro)
- E) Sulfamethoxazole/trimethoprim

89. A 5-year-old male is brought to the emergency department (ED) by his parents for evaluation after falling from a kitchen table just prior to arrival. The mother states that the child climbed onto the table and fell off before she could reach him. He landed directly on the right side of his head and did not lose consciousness. She witnessed 1 episode of vomiting after the fall but no additional episodes since then. On physical examination, the patient is tearful but consolable. You note a right-sided parietal hematoma. There are no palpable skull fractures or signs of altered mental status.

Which one of the following would be the most appropriate next step in management?

- A) Providing reassurance and recommending outpatient follow-up
 - B) Observation in the ED for worsening symptoms
 - C) Radiography of the skull
 - D) CT of the head and neck
 - E) Contacting child protective services
90. A 32-year-old male presents for advice on smoking cessation. He has smoked 10 cigarettes per day for the past 10 years. His grandmother was recently diagnosed with lung cancer, and he is motivated to quit. You advise him that smoking cessation is the most important action that he can take for his health, and you counsel him about effective behavioral interventions.

Which one of the following behavioral changes would improve this patient's chances of smoking cessation?

- A) Creating a smoke-free home
 - B) Increasing his caffeine intake
 - C) Switching to menthol cigarettes
 - D) Undergoing hypnotherapy
91. Which one of the following medications is the recommended initial management of COPD in patients who are classified as GOLD group A?
- A) Inhaled umeclidinium (Incruse Ellipta)
 - B) Inhaled aclidinium bromide/formoterol (Duaklir Pressair)
 - C) Inhaled budesonide/formoterol (Symbicort)
 - D) Inhaled fluticasone furoate/umeclidinium/vilanterol (Trelegy Ellipta)
 - E) Nebulized formoterol (Perforomist)

92. A 29-year-old female with type 2 diabetes is brought to the emergency department by her family, who state that she has grown increasingly confused over the last few days, with additional symptoms of polyuria and polydipsia. Her initial vital signs demonstrate hypotension and tachycardia.

Laboratory Findings

Serum glucose	750 mg/dL
Serum osmolality	350 mOsm/kg H ₂ O (N 275–295)
Urine ketones	trace
Arterial pH	7.3
Anion gap	normal

Which one of the following would be the most appropriate initial treatment?

- A) Subcutaneous short-acting insulin
 - B) Subcutaneous long-acting insulin
 - C) Intravenous isotonic fluid hydration
 - D) Intravenous isotonic fluid hydration with sodium bicarbonate
93. A 65-year-old male with symptomatic heart failure presents for follow-up. He currently has no symptoms of dyspnea at rest, but has moderate to significant limitations with ordinary activity. His most recent echocardiogram showed a left ventricular ejection fraction of 38%. His last hemoglobin A_{1c} was the highest it has ever been at 6.3%, and he does not take any glucose-lowering medications. His current medications include losartan (Cozaar), carvedilol (Coreg), furosemide (Lasix), and spironolactone (Aldactone). On examination he has a blood pressure of 120/75 mm Hg and an oxygen saturation of 94% on room air, and there are no pulmonary crackles or peripheral edema.

Adding which one of the following would be most likely to further reduce this patient's mortality risk?

- A) A calcium channel blocker
- B) A GLP-1 receptor agonist
- C) An SGLT2 inhibitor
- D) A thiazide diuretic
- E) Metformin

94. A 45-year-old male with a past medical history of hyperlipidemia and obesity presents with concerns of polydipsia and polyuria that have progressively worsened over the last 2 weeks. His older sister was recently diagnosed with type 2 diabetes and your patient asks to be tested.

Which one of the following test results would be sufficient to diagnose diabetes in this patient?

- A) A hemoglobin A_{1c} $\geq 6.0\%$
 - B) A 1-hour plasma glucose level ≥ 140 mg/dL on 50-g oral glucose testing
 - C) A fasting plasma glucose level ≥ 120 mg/dL
 - D) A random blood glucose level ≥ 200 mg/dL
95. A 36-year-old male presents to your office with continued pain in his right arm after sustaining a right forearm fracture 6 weeks ago that was treated with cast immobilization. At the time of the injury, he noted a severe, throbbing pain in his right forearm that improved with NSAID therapy. Now, he feels a severe, sharp pain with burning and numbness throughout the arm and hand that worsens with any pressure, including clothing. He reports weakness in his right hand and notes that he drops objects such as cups. He is afebrile and his vital signs are within normal limits. An examination reveals pain to light touch in the right arm. The right arm is warmer to the touch than the left arm with trace edema noted. The patient's right-hand grip is 4/5, and his left-hand grip is 5/5. You note 2+ radial and brachial pulses and normal bilateral reflexes.

Which one of the following is the most likely etiology of this patient's pain?

- A) Cellulitis
- B) Complex regional pain syndrome
- C) Factitious disorder
- D) Functional neurologic symptom disorder (conversion disorder)
- E) Peripheral neuropathy

96. A 28-year-old nulligravida has a 2- to 3-day history of severe lower abdominal pain. On examination the patient's temperature is 37.7 °C (99.9 °F), bowel sounds are normal, and there is diffuse lower abdominal tenderness without rebound. On pelvic examination there is adnexal tenderness and pain on cervical motion. Vaginal discharge shows leukocytes. A pregnancy test is negative.

In addition to sexually transmitted infection testing and empiric treatment, which one of the following should you recommend?

- A) No further testing
 - B) A repeat pregnancy test in 7 days
 - C) Pelvic ultrasonography
 - D) Pelvic MRI
 - E) Laparoscopy
97. A 65-year-old male with a past medical history of hypertension and hyperlipidemia presents to the urgent care clinic approximately 20 hours after experiencing an episode of sudden-onset right-hand weakness. He was suddenly unable to grasp his utensils while eating dinner. These symptoms resolved after about 20 minutes. His wife had encouraged him to seek urgent evaluation, but he declined because his symptoms had resolved.

Today the patient's vital signs include a blood pressure of 124/72 mm Hg, a heart rate of 80 beats/min, and a respiratory rate of 12/min. The neurologic examination findings are normal. An EKG shows normal sinus rhythm. The ABCD² score is 4.

Further management includes which one of the following regimens?

- A) Aspirin, 81 mg daily
- B) Clopidogrel (Plavix), 300 mg on day 1, followed by 75 mg daily
- C) Clopidogrel, 300 mg on day 1, followed by 75 mg daily on days 2–10; plus aspirin, 81 mg daily
- D) Clopidogrel, 300 mg on day 1, followed by 75 mg daily on days 2–21; plus aspirin, 81 mg daily

98. A 55-year-old patient with a previous history of opioid use disorder and untreated hypertension presents to your office to establish care. They are now in recovery and would like to catch up on screening laboratory studies. Their blood pressure today is 160/90 mm Hg. Laboratory studies show elevated AST, ALT, and alkaline phosphatase levels and are positive for hepatitis C antibody and hepatitis C viral load. Testing for hepatitis B and HIV is negative. Lipids are elevated. You calculate an atherosclerotic cardiovascular disease 10-year risk score of 10%. A right upper quadrant ultrasound shows fibrotic changes of the liver. The patient is surprised that they have liver damage, as they feel well and have never consumed much alcohol.

In addition to referring the patient to a gastroenterologist, which one of the following should you begin now to improve fibrotic changes?

- A) Treatment of hepatitis C
 - B) Prophylactic antibiotics
 - C) A nonselective β -blocker
 - D) Pre-exposure prophylaxis (PrEP) for HIV
 - E) Upper endoscopy with banding of any varices
99. A 46-year-old female sees you for an annual well woman visit. Her only concern is pelvic pressure that becomes more bothersome throughout the day. She notes occasional urinary incontinence of small amounts when coughing or running. She says these symptoms mildly bother her, and mostly wants to make sure there is not a more serious problem. On physical examination, using the lower half of a bivalve speculum, you note a prolapse of the anterior wall, 0.5 cm proximal to the hymenal ring.

Which one of the following would be the most appropriate next step for this patient?

- A) Reassurance only
- B) Pelvic floor muscle training
- C) Topical estrogen
- D) Surgical consultation

100. A 72-year-old male with a past medical history of coronary artery disease, COPD, and hyperlipidemia sees you because of an acute onset of left lower leg pain 1 hour ago. He was in his usual state of health when the pain occurred. His left foot is numb and cold, and he is having difficulty walking. He has a 60-pack-year smoking history. Physical examination findings include a mottled appearance of the left lower leg and foot. Posterior tibial and dorsalis pedis pulses are absent.

Which one of the following would be the most appropriate first step in management?

- A) Rivaroxaban (Xarelto)
- B) Stat duplex ultrasonography
- C) Stat CT angiography
- D) Expedited outpatient referral to a vascular surgeon
- E) Transfer to the nearest emergency department with access to a vascular surgeon

101. A 22-year-old male presents with a 5-day history of left lower leg pain over the midshaft of his tibia. Two months ago, he started training for a half marathon. Radiographs performed yesterday at an urgent care clinic were normal. You suspect a bone stress injury.

Which one of the following is the preferred definitive imaging modality to evaluate for a bone stress injury?

- A) Repeat radiography in 1 week
- B) Point-of-care ultrasonography
- C) CT
- D) MRI
- E) Nuclear bone scintigraphy

102. A 32-year-old female presents to an urgent care clinic with a flare-up of her usually well-controlled chronic urticaria. She takes cetirizine (Zyrtec), 10 mg daily.

Which one of the following should you recommend?

- A) A pseudoallergen-free diet
- B) Doubling the cetirizine dosage
- C) A nasal corticosteroid
- D) Cyclosporine (Sandimmune)
- E) Omalizumab (Xolair)

103. A 35-year-old female comes to your office with an 8-day history of nasal congestion, rhinorrhea, and cough. Based on guidelines from the Infectious Diseases Society of America, this patient should be treated with antibiotic therapy if she has which one of the following additional symptoms?

- A) A fever up to 38.0 °C (100.4 °F)
- B) Halitosis and tooth pain
- C) An increase in nasal discharge after an initial improvement of symptoms
- D) Purulent rhinorrhea

104. You are seeing a 44-year-old female who has a 25-pack-year smoking history. She quit smoking 6 years ago after a friend received a diagnosis of lung cancer.

According to U.S. Preventive Services Task Force recommendations, lung cancer screening should be initiated at what age in this patient?

- A) No screening is indicated
- B) 45
- C) 50
- D) 55
- E) 60

105. A 25-year-old male presents with a 5-day history of dysuria and mucopurulent penile discharge. On examination the urethral meatus is erythematous and a small amount of discharge is noted. A urine dipstick is positive for leukocyte esterase and a nucleic acid amplification test is negative for *Neisseria gonorrhoeae* and *Chlamydia trachomatis*.

Which one of the following would be the most appropriate treatment?

- A) Azithromycin (Zithromax), 1 g orally once
- B) Levofloxacin, 500 mg orally for 7 days
- C) Ceftriaxone, 500 mg intramuscularly once
- D) Gentamicin, 240 mg intramuscularly once, plus azithromycin, 2 g orally once
- E) Metronidazole, 2 g orally once, plus doxycycline, 100 mg orally twice daily for 7 days

106. A 55-year-old female presents to the emergency department with what she describes as the worst headache of her life that began suddenly 1 hour ago. Her medical history is significant for tobacco use and uncontrolled hypertension.

She indicates that she has had a few headaches over the last month, but today's headache came on suddenly and is extremely intense, unlike the previous headaches. There is no history of head trauma. Her accompanying relative adds that the patient has been experiencing waxing and waning levels of consciousness since the sudden onset of the headache. She has vomited several times, secondary to pain from the headache.

Which one of the following is the most important, immediate next step in management?

- A) Morphine
- B) Ondansetron
- C) Head CT
- D) Brain MRI
- E) Lumbar puncture

107. A 28-year-old male who uses cannabis daily presents with a 2-week history of persistent nausea and vomiting. He notes little relief from previously prescribed ondansetron but does feel better during a hot shower.

While tapering his cannabis use, which one of the following would be most likely to improve this patient's nausea?

- A) Topical diclofenac gel
- B) Haloperidol
- C) Metoclopramide (Reglan)
- D) Prednisone
- E) Prochlorperazine

108. A 30-year-old female with fibromyalgia sees you for follow-up. Her symptoms include widespread pain and prominent fatigue. At your advice, she has engaged in a trial of multidisciplinary, nonpharmacologic treatment including structured exercise, cognitive behavioral therapy, and therapeutic massage, and has experienced only partial improvement in her symptoms. She expresses interest in trying a medication to treat her pain.

Which one of the following medications would be most likely to improve this patient's pain?

- A) Acetaminophen
- B) Duloxetine (Cymbalta)
- C) Gabapentin (Neurontin)
- D) Hydrocodone/acetaminophen
- E) Ibuprofen

109. An 8-year-old male is brought to your office by his parents because of concerns about ongoing nocturnal enuresis. He wets his bed several nights per week despite behavioral interventions. He has no daytime urinary symptoms. Physical examination findings and urinalysis are normal. You discuss treatment options and the parents indicate a preference for medication, as the patient will be attending an overnight summer camp in several weeks.

Which one of the following would be the preferred pharmacologic treatment for this child's nocturnal enuresis?

- A) Desmopressin (DDAVP)
- B) Imipramine
- C) Oxybutynin
- D) Tamsulosin (Flomax)

110. A 25-year-old male with a childhood diagnosis of attention-deficit/hyperactivity disorder (ADHD) presents for re-evaluation of ADHD. He did not take medication for ADHD as a child. He is now having problems with attention and concentration that are affecting his work. He was recently placed on a stimulant medication, but had to discontinue it when his blood pressure and heart rate became elevated. He is still interested in pharmacologic therapy.

Which one of the following would be the most appropriate next medication?

- A) Atomoxetine (Strattera)
 - B) Clonidine
 - C) Escitalopram (Lexapro)
 - D) Guanfacine
 - E) Nortriptyline (Pamelor)
111. A 19-year-old college basketball player consults you for follow-up after sustaining an anterior shoulder dislocation for the first time. He initially presented to the emergency department where his shoulder was reduced and immobilized in a sling.

Which one of the following is most likely to decrease the risk of future recurrence?

- A) Continued sling use for 4 weeks
 - B) Completion of a course of physical therapy
 - C) Administration of an intra-articular corticosteroid injection
 - D) Surgical repair
112. A 32-year-old gravida 2 para 2 female who is 3 weeks post partum presents to the local urgent care center for left breast pain, swelling, and redness that started a few hours ago. She does not have fever or chills. A breast examination is significant for warm, tender, and mildly erythematous nodules localized to the upper outer quadrant of the left breast.

In addition to initiating NSAIDs, which one of the following would be the most appropriate next step in management?

- A) Applying cold compresses to the affected breast
- B) Applying warm compresses to the affected breast
- C) Using a manual breast pump on the affected breast for complete breast emptying
- D) Obtaining a CBC and C-reactive protein levels
- E) Obtaining a breast milk culture

113. A young couple in your practice presents for preconception counseling. Both partners are known carriers of cystic fibrosis (CF). They ask what risk this poses to their future children.

Which one of the following accurately reflects the risk of CF disease transmission to any potential offspring of this couple?

- A) 25% risk of being affected and 25% risk of being a carrier
 - B) 25% risk of being affected and 50% risk of being a carrier
 - C) 50% risk of being affected and 25% risk of being a carrier
 - D) 50% risk of being affected and 50% risk of being a carrier
114. A 47-year-old nurse presents with a 9-month history of fatigue and brain fog that has kept them from working. Their symptoms began after a confirmed diagnosis of COVID-19. They report sleeping at least 7 hours per night but wake unrefreshed and experience lightheadedness throughout the day. They used to run 4 miles 5 days per week but now report feeling exhausted and experience energy crashes after showering and getting dressed. The patient's vital signs and BMI are normal. A physical examination is unremarkable. Their Patient Health Questionnaire–9 (PHQ-9) score is 3. A CBC, TSH level, basic metabolic panel, and erythrocyte sedimentation rate are within the normal range.

Which one of the following is the most likely diagnosis?

- A) Depression
- B) Fibromyalgia
- C) Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS)
- D) Obstructive sleep apnea
- E) Systemic lupus erythematosus

115. A 62-year-old male is brought to the clinic by his wife for fatigue. His wife says he has been sleeping in the recliner rather than in bed and has been less active over the last several weeks. He reports that his clothes have been fitting tightly and he has gained weight despite feeling full early in a meal. He also notes difficulty bending over to tie his shoes and pet the dog. He has no chest pain or pressure and has not traveled recently. His past medical history is notable for a myocardial infarction 5 years ago treated with a stent, and elevated cholesterol. He has a 30-pack-year smoking history but quit 20 years ago. His medications include aspirin, 81 mg daily; lisinopril (Zestril), 20 mg daily; and metoprolol succinate (Toprol-XL), 50 mg daily.

On examination he has a blood pressure of 110/65 mm Hg, a pulse rate of 89 beats/min, and an oxygen saturation of 95% on room air. You also note mildly increased work of breathing, a regular heart rate, and a laterally displaced apical impulse.

Which one of the following tests would most likely confirm this patient's underlying diagnosis?

- A) EKG
 - B) Echocardiography
 - C) X-ray of the chest
 - D) CT angiography of the heart
 - E) CT angiography of the lungs
116. A 3-year-old male is brought to your office by his parents for a routine well child visit. He is up to date on vaccines and the parents have no concerns to discuss today.

According to the U.S. Preventive Services Task Force, this child should be screened for which one of the following conditions?

- A) Amblyopia
- B) Anemia
- C) Autism spectrum disorder
- D) Dental caries
- E) Hearing loss

117. A 50-year-old female comes to your clinic for an annual wellness examination. Her past medical history is notable for dermatomyositis diagnosed 2 years ago. She is otherwise healthy and her BMI is normal.

Patients with dermatomyositis are at increased risk of which one of the following conditions?

- A) Chronic kidney disease
 - B) Diabetes mellitus
 - C) Hyperlipidemia
 - D) Malignancy
 - E) Obesity
118. Of the following cognitive biases, which one best describes the practice of prioritizing data that supports an initial impression, even when the initial impression is incorrect?
- A) Anchoring
 - B) Availability
 - C) Implicit
 - D) Outcomes
119. An otherwise healthy 15-year-old female presents for evaluation of a 7-month history of heavy menstrual bleeding. Her periods are 10 days long, irregular, and require 10 tampons per day. She sometimes passes large clots. Other than some fatigue, she feels well today and her vital signs are normal. A point-of-care hemoglobin level is 10.2 g/dL (N 12.0–16.0).

Which one of the following would be the most appropriate next step in evaluation?

- A) Testing for a bleeding disorder
- B) FSH and LH levels
- C) Estrogen and progesterone levels
- D) Pelvic ultrasonography
- E) Pelvic MRI

120. A 45-year-old patient presents with a 6-month history of epigastric burning pain that occurs several days per week. The pain does not seem to have a clear correlation to eating. There has been no substernal pain or regurgitation and no weight loss, vomiting, diarrhea, constipation, fever, night sweats, or fatigue. The patient has no prior medical history and takes no medications, but has tried calcium carbonate occasionally without benefit. A physical examination reveals normal vital signs and no abdominal abnormalities or enlarged lymph nodes.

Which one of the following would be the most appropriate next step?

- A) Sucralfate (Carafate)
- B) Stool antigen testing for *Helicobacter pylori*
- C) Serum testing for *H pylori*
- D) Abdominal ultrasonography
- E) Referral for upper endoscopy

121. A 48-year-old female comes to your office for a health maintenance examination. She is currently asymptomatic but has a medical history of renal calculi. Laboratory studies reveal a serum calcium level of 12.1 mg/dL (N 8.7–10.2), confirmed by an ionized calcium level.

Which one of the following would be the most appropriate next step at this time?

- A) A 24-hour urinary calcium level
- B) A parathyroid hormone level
- C) A vitamin D level
- D) CT of the chest, abdomen, and pelvis

122. A 35-year-old female presents to your office with a 3-day history of right-sided facial weakness. She has flattening of the nasolabial fold and is unable to elevate the right corner of her mouth. She is able to close her right eye and raise her eyebrows.

Which one of the following would be the most appropriate next step in management?

- A) Reassurance only
- B) Prednisone plus valacyclovir (Valtrex)
- C) MRI of the brain
- D) Plasmapheresis
- E) Surgical decompression

123. A previously healthy 70-year-old yoga instructor is brought to the emergency department (ED) by her students after falling while practicing yoga. Her students report that she struck her head on the floor and was unresponsive for a few seconds, but recovered quickly. She was slightly confused after the injury, but was able to stand and describe the incident. On initial assessment in the ED, she is holding her head in obvious discomfort and a hematoma is developing on her forehead. She is alert and oriented, with a Glasgow Coma Scale score of 15. On examination she is neurovascularly intact. Her vital signs are normal except for an elevated heart rate of 110 beats/min and a blood pressure of 150/92 mm Hg.

Which one of the following would be the most appropriate next step in the management of this patient's injury?

- A) Discharging home with concussion precautions
 - B) Observation in the ED, and acetaminophen
 - C) Observation in the ED, and ibuprofen
 - D) CT of the head
124. A 3-day-old infant presents to your clinic for a routine check-up. The parents report that the infant has been feeding well but has had fewer wet diapers than expected. On examination, the infant appears slightly lethargic with dry skin. You need to assess the hydration status of the infant.

Which one of the following would be the most reliable clinical sign for assessing dehydration in this 3-day-old infant?

- A) Capillary refill time
- B) Decreased urine output
- C) Dry mucous membranes
- D) Skin turgor
- E) Sunken fontanelle

125. A 16-year-old female presents with her parents because she has been experiencing a loss of interest in her usual activities, trouble sleeping, difficulty concentrating, and low energy over the last month. After a discussion with the patient and her parents, you refer her for cognitive behavioral therapy and start an antidepressant.

Which one of the following medications is approved by the FDA to treat depression in patients such as this?

- A) Esketamine nasal spray (Spravato)
- B) Bupropion
- C) Fluoxetine (Prozac)
- D) Venlafaxine

126. A 35-year-old patient presents with a 14-day history of a productive cough and green sputum. They report they initially had a fever but that it resolved after a few days. The cough is interfering with their sleep and daily activity, and they have tried cough drops without relief. They have a history of hypothyroidism for which they take levothyroxine (Synthroid), and do not have a history of asthma. They do not smoke, and their immunizations are up to date.

On examination, the patient is afebrile and has an oxygen saturation of 98% on room air, a heart rate of 82 beats/min, and a blood pressure of 120/82 mm Hg. A lung examination reveals mild rhonchi that clear with cough, and you do not note any wheezing or egophony.

Which one of the following treatments would be the most effective in reducing this patient's cough?

- A) An albuterol inhaler (Proventil, Ventolin)
- B) Amoxicillin/clavulanate (Augmentin)
- C) Azithromycin (Zithromax)
- D) Codeine/guaifenesin syrup
- E) Dextromethorphan (Delsym)

127. A 20-year-old male presents to the urgent care center for left ankle pain and swelling following an ankle inversion injury that occurred a few hours earlier. A physical examination reveals mild swelling of the left ankle joint.

Using the Ottawa foot and ankle rules, which one of the following findings, if present, would be an indication for radiography for an acute ankle inversion injury?

- A) Bone tenderness at the anterior edge of the lateral malleolus
 - B) Bone tenderness at the anterior edge of the medial malleolus
 - C) Bone tenderness at the base of the fourth metatarsal
 - D) Bone tenderness of the navicular bone
 - E) Inability to bear weight on the left foot for at least 10 steps after the injury
128. A generally healthy 13-year-old male presents with his parents because they are concerned that he has very large-volume stools that clog the toilet. The patient says that he averages fewer than 2 bowel movements per week and they are sometimes painful. He does not have any abdominal pain and has not seen any blood in his stools. Examination findings are unremarkable. His height, weight, and BMI percentiles have remained steady for years. A CBC, comprehensive metabolic panel, ferritin level, and TSH level are all normal. An abdominal x-ray is normal.

Which one of the following would be the most appropriate next step?

- A) Polyethylene glycol (MiraLAX) daily
- B) Probiotics daily
- C) Magnetic resonance enterography
- D) Colonoscopy

129. A 45-year-old male sees you for routine follow-up of diabetes mellitus. He smokes 1 pack of cigarettes per day and does not have any other medical conditions. His diabetes is well controlled on liraglutide (Victoza) and metformin. He has an LDL-cholesterol level of 120 mg/dL, an HDL-cholesterol level of 55 mg/dL, and a triglyceride level of 300 mg/dL. His 10-year atherosclerotic cardiovascular disease risk is 8.2%.

According to 2018 guidelines from the American College of Cardiology/American Heart Association, the most appropriate recommendation for this patient would be

- A) atorvastatin (Lipitor), 40 mg daily
 - B) ezetimibe (Zetia), 10 mg daily
 - C) fenofibrate, 160 mg daily
 - D) pravastatin, 20 mg daily
 - E) simvastatin (Zocor), 40 mg daily
130. A 37-year-old male comes to the primary care clinic with a several-month history of worsening paresthesia in his right foot. He says that he sometimes has numbness and a hot sensation in his feet and has difficulty using the pedals of his car. Sometimes, he has pain that wakes him up at night but eases when he walks around the room.

He has no significant past medical history but did sustain a severe right ankle sprain about 1 year ago. His vital signs are unremarkable. Examination of the ankle reveals pain along the medial malleolus and a positive Tinel sign at the ankle. An x-ray of the ankle is unremarkable.

Which one of the following is the most likely diagnosis?

- A) Achilles tendinopathy
- B) Haglund deformity
- C) Heel pad syndrome
- D) Plantar fasciitis
- E) Tarsal tunnel syndrome

131. A 36-year-old female comes to your office for a health maintenance examination. She mentions that she engages in amateur rifle target shooting matches.

Which one of the following would be the most appropriate advice concerning this patient's physical safety?

- A) She should wear electronic hearing protection that amplifies quiet sounds
- B) She should wear noise-canceling headphones
- C) She should wear loose-fitting clothing
- D) She should wear her regular glasses so that she has eye protection

132. A 66-year-old male presents to your office with a 2-hour history of discomfort in the midsternal area that began while taking a walk earlier today. He says that he feels 90% better. He had a Welcome to Medicare visit 1 year ago, in which an EKG was normal. You order an EKG today.

Which one of the following EKG findings, if present in this patient, should prompt urgent transport to the emergency department?

- A) First-degree atrioventricular block
- B) Frequent PVCs
- C) Left bundle branch block
- D) Sinus tachycardia
- E) Voltage criteria for left ventricular hypertrophy

133. A generally healthy 56-year-old female sees you for follow-up of pneumonia diagnosed in the emergency department (ED). A chest x-ray performed in the ED showed a left lower lobe infiltrate and several compression fractures involving the thoracic vertebrae. A DXA scan performed 7 months ago showed a T-score of -1.9 . Her daily medications include a multivitamin; calcium, 1000 mg; and vitamin D₃ (cholecalciferol), 800 IU. Today she feels well and has no concerns.

This patient would be most likely to benefit from therapy for which one of the following conditions?

- A) Osteopenia
- B) Osteoporosis
- C) Spondylolisthesis
- D) Spondylolysis

134. A gravida 2 para 1 at 28 weeks' gestation presents with an episode of second-trimester bleeding that spontaneously resolved. She has not had any prenatal laboratory studies during either of her pregnancies. Ultrasonography today shows a healthy fetus and a small placental abruption.

Which one of the following would be most important to order at this time?

- A) An antiphospholipid antibody profile
- B) An HIV test
- C) An Rh factor test
- D) A toxoplasmosis antibody test
- E) A urine toxicology screen for cocaine

135. A generally healthy 50-year-old nurse presents to your clinic 1 day after sustaining an accidental needlestick containing the blood of a person with active hepatitis B virus infection who has not yet begun treatment. Your patient completed the hepatitis B (HepB) vaccine series at age 30 before beginning his first nursing job. Hepatitis serologies drawn yesterday at his place of employment were positive for hepatitis B surface antibody (>10 mIU/mL).

Which one of the following would be recommended for this patient?

- A) No additional vaccination
- B) Hepatitis B immune globulin (HBIG) only
- C) Recombinant HepB vaccine only
- D) Both HBIG and recombinant HepB vaccine

136. A 20-year-old female presents for a routine health maintenance examination and has no medical concerns today. She does not have a history of sexual activity, and screenings for alcohol and substance use, intimate partner violence, anxiety, depression, and suicidality are negative. An examination reveals fair skin, an elevated BMI of 31 kg/m², and normal blood pressure.

According to the U.S. Preventive Services Task Force, you should recommend screening for which one of the following?

- A) Cervical cancer
 - B) Chlamydia
 - C) Diabetes mellitus
 - D) HIV
 - E) Skin cancer
137. A 65-year-old male presents to your clinic with symptoms of flashing lights and increased floaters in his left eye. He does not have any pain, and he has no past medical history. His vital signs are normal. His visual acuity is at baseline when wearing his glasses, and you note a visual field defect on the left.

Which one of the following is the most likely diagnosis?

- A) Migraine with aura
- B) Central retinal artery occlusion
- C) Retinal detachment
- D) Globe rupture
- E) Acute ischemic stroke

138. A 46-year-old male presents for an acute visit due to a 2-week history of low back pain. He was moving furniture when the pain started, and he felt a pop in his right lower back area. The pain is severe enough that it keeps him from standing up straight and is causing him trouble with sleeping. He does not have fever or chills, numbness, paresthesia, or weakness in his lower extremities. He has no radicular pain on a straight leg-raising test. He went to a chiropractor who did x-rays and told him he needed to get an MRI.

Which one of the following, if present, would be an indication for advanced imaging?

- A) Current cigarette use
 - B) Current intravenous drug use
 - C) Disrupted sleep due to nocturnal pain
 - D) Restricted posture due to pain
 - E) New urinary retention
139. How long after intranasal administration does naloxone (Narcan) reach peak plasma concentrations?
- A) Immediately
 - B) 30–60 seconds
 - C) 3–5 minutes
 - D) 15–30 minutes
 - E) 1–2 hours
140. A 24-year-old male presents for evaluation of a bulge in his groin. He reports that it is larger at the end of the day and with straining but says it often goes away when he lies down. He reports a mild dull ache that limits his ability to lift lumber at work. On examination there is no obvious bulge in the inguinal area. When the patient bears down, a bulge becomes apparent.

Which one of the following is the most appropriate next step?

- A) Inguinal ultrasonography
- B) Scrotal ultrasonography
- C) Abdominal and pelvic CT
- D) Physical therapy
- E) Surgical referral

141. A 72-year-old male presents to your office with his daughter who is concerned about her father's recent decline in memory. His history is notable for hypertension, urinary urgency with occasional incontinence, and GERD. On examination, you notice a slow, wide-based gait. A Mini-Mental State Examination (MMSE) score is 22/30. MRI of the brain is notable for diffuse dilation of the lateral, third, and fourth ventricles, and the diagnosis is confirmed with a high-volume lumbar puncture.

Which one of the following is the most appropriate treatment for this condition?

- A) Carbidopa/levodopa (Sinemet), 12.5/50 mg daily
- B) Donepezil (Aricept), 5 mg daily
- C) Levothyroxine (Synthroid), 50 µg daily
- D) Ventriculoperitoneal shunt
- E) Surgery with postoperative chemotherapy

142. A 65-year-old male presents to the emergency department with an acute onset of bloody urine confirmed to be hematuria on urinalysis. He does not have any pain, fever, or recent trauma. His medical history is significant for hypertension that is managed with lisinopril (Zestril). A physical examination is unremarkable.

Which one of the following would be the most appropriate next step in evaluation?

- A) Repeating the urinalysis in 1 month
- B) Initiating empiric antibiotic therapy and re-evaluating in 1 week
- C) Ordering urine cytology
- D) Referring the patient for immediate cystoscopy and imaging studies

143. A 40-year-old male presents to establish care. He smokes 1½ packs of cigarettes per day and is uncertain about cutting back or quitting. While planning to enact a patient-centered strategy to help him overcome his reluctance, you recall that evidence supports the application of a counseling technique that uses open-ended questions, affirmations, reflective listening, and summary statements to help foster a sense of patient autonomy and effect positive behavioral change.

Which one of the following counseling strategies best fits this description?

- A) BATHE (background, affect, troubles, handling, empathy)
- B) The five A's (ask, advise, assess, assist, arrange)
- C) Motivational interviewing
- D) Transtheoretical (stages of change) model

144. A 35-year-old female with type 1 diabetes presents with a skin lesion on her leg. The lesion began as a small, dusky red, elevated nodule with a sharply circumscribed border, and slowly enlarged to become the lesion shown below.

The most likely diagnosis is



- A) diabetic dermopathy
 - B) diabetic lipoatrophy
 - C) diabetic rubeosis
 - D) eruptive xanthoma
 - E) necrobiosis lipoidica diabetorum
145. Metformin can cause which one of the following?

- A) Chromatopsia
- B) Ileus
- C) Renal failure
- D) Vitamin B₁₂ deficiency
- E) Weight gain

146. A 3-year-old female is brought to your office by her parents with fever, cough, and decreased appetite. On examination you hear crackles in the left upper lobe and a chest x-ray confirms a left upper lobe consolidation. She appears well hydrated and her vital signs are normal. On examination she is drinking water and interacting normally.

Which one of the following would be the most appropriate antibiotic for this patient's condition?

- A) Amoxicillin
 - B) Azithromycin (Zithromax)
 - C) Cefdinir
 - D) Doxycycline
 - E) Sulfamethoxazole/trimethoprim
147. A 35-year-old male presents to the urgent care clinic with pain in his right index finger after jamming it during basketball practice. An examination demonstrates a gross deformity of the proximal interphalangeal (PIP) joint of the affected finger. There is dorsal angulation of the proximal phalanx. Capillary refill, sensation, and pulses are all intact. Because the patient has pain with attempted movement of the finger, a digital block using 1% lidocaine without epinephrine is used to complete the examination. Complete x-rays, including true lateral films, reveal a very small fracture of the middle phalanx involving a minimal (<20%) area of the articular surface.

Which one of the following would be the most appropriate next step in the treatment of this patient's finger dislocation?

- A) Discharging home, and repeating x-rays in 1 week
- B) Reduction, followed by splinting in a flexed position and buddy taping
- C) MRI of the finger
- D) Referral for physical therapy with instructions to start early range-of-motion exercises
- E) Urgent referral to an orthopedist for hand surgery

148. A 27-year-old female comes to your office because she is having moderately severe, diffuse abdominal pain with nausea and approximately 5 loose, greasy stools daily. She recently returned from a 2-week backpacking trip in Nicaragua. Stool studies are positive for *Giardia*.

Which one of the following would be the most appropriate treatment?

- A) Supportive care only
- B) Azithromycin (Zithromax)
- C) Ciprofloxacin (Cipro)
- D) Ivermectin (Stromectol)
- E) Metronidazole (Flagyl)

149. A 49-year-old female with a past medical history of epilepsy presents to your office because of sadness, sleep disruption, generalized anhedonia, and a recent increase in appetite. Her Patient Health Questionnaire–9 (PHQ-9) score is elevated at 14.

Which one of the following medications should be AVOIDED in this patient?

- A) Bupropion
- B) Duloxetine (Cymbalta)
- C) Fluoxetine (Prozac)
- D) Mirtazapine (Remeron)
- E) Sertraline (Zoloft)

150. A 20-year-old female presents for a health maintenance examination. She has no significant past medical history and no concerns.

Based on U.S. Preventive Services Task Force guidelines, screening for which one of the following would be most appropriate at this time?

- A) Hepatitis C
- B) Hyperlipidemia
- C) Cervical cancer
- D) Skin cancer

151. A 75-year-old male with acute confusion presents to the emergency department and is found to have a blood pressure of 280/140 mm Hg. Which one of the following potential etiologies would provide the strongest indication for lowering his systolic blood pressure to a target of 120 mm Hg within 20 minutes?

- A) Acute aortic dissection
- B) Acute ischemic stroke
- C) Acute renal failure
- D) Hypertensive encephalopathy
- E) Intracerebral hemorrhage

152. A 56-year-old male sees you because he has been falling asleep at work in the afternoon. Upon questioning he notes that he is very tired all the time. Although he is in bed 8 hours per night and wakes with an alarm, he gets up 3 times nightly to urinate. He says that he has been told he snores. He does not have any constipation, dry skin, mood changes, stomach pain, changes in urine stream, shortness of breath, or chest pain.

The patient is 173 cm (68 in) tall and weighs 95 kg (209 lb). His vital signs include a blood pressure of 140/88 mm Hg, a pulse rate of 79 beats/min, and a respiratory rate of 16/min.

Which one of the following is most likely to yield the correct diagnosis?

- A) A serum TSH level
- B) Vitamin B₁₂ and folate levels
- C) Iron studies
- D) A comprehensive metabolic panel
- E) Polysomnography

153. A 16-year-old female presents for a routine well child examination. She has never been sexually active and does not use illicit substances.

Based on guidelines from the U.S. Preventive Services Task Force, screening patients such as this for which one of the following has the strongest evidence of benefit?

- A) Anxiety
- B) Chlamydia and gonorrhea
- C) Eating disorders
- D) Elevated blood pressure
- E) Scoliosis

154. A 50-year-old female presents for evaluation of unilateral shoulder pain without a history of acute injury. She has tried ibuprofen with minimal symptomatic relief. Plain radiographs of the affected shoulder taken at an urgent care clinic last week demonstrated no abnormal findings.

Which one of the following physical examination findings would best support a diagnosis of adhesive capsulitis as the primary etiology of this patient's shoulder pain?

- A) Apprehension in a 90° position with abduction and external rotation
 - B) Crepitus on passive abduction with preserved passive range of abduction
 - C) Restricted active range of abduction with preserved passive range of abduction
 - D) Restricted passive range of abduction and external rotation
 - E) Weakness on resisted abduction and external rotation
155. A 60-year-old male with a history of type 2 diabetes presents to your office because he is concerned about an ulcer on the bottom of his right foot. He is uncertain how long the ulcer has been present. He is afebrile and otherwise asymptomatic. His most recent hemoglobin A_{1c} is 9.7% and was obtained 9 months ago. His medications include metformin, 2000 mg daily, and insulin glargine (Lantus), 34 units nightly. He does not regularly check his blood glucose levels.

On examination, the patient has an ulcer on the plantar surface of the right foot overlying the head of the fifth metatarsal, measuring 1.0 cm in diameter. The ulcer appears deep with erythema that extends approximately 2.5 cm from the margin of the ulcer. A probe-to-bone test is positive.

Which one of the following would be the most appropriate next test to evaluate for osteomyelitis?

- A) A WBC count
- B) A culture swab
- C) Transcutaneous oximetry
- D) Plain radiography

156. A 63-year-old female with hypertension sees you for routine follow-up. She has been bothered by swelling in her legs for the last several months. Her blood pressure is well controlled and she does not have any shortness of breath, cough, difficulty breathing while lying down, or pain in her legs when walking. She thinks the swelling is better earlier in the day. Her current medications include hydrochlorothiazide, lisinopril (Zestril), estrogen, levothyroxine (Synthroid), and as-needed acetaminophen. A physical examination demonstrates mild pitting edema of both legs from the ankle to the distal third of each calf. There are 2+ pedal pulses and no skin changes. Examinations of the heart and lungs are normal.

Which one of the following would be the most appropriate next step?

- A) Holding estrogen, and monitoring for improvement in peripheral edema
- B) Holding lisinopril, and monitoring for improvement in peripheral edema
- C) Bilateral ankle-brachial indices
- D) Bilateral venous compression ultrasonography
- E) Bilateral venous duplex ultrasonography with reflux

157. Which one of the following antibiotics would be recommended for the treatment of syphilis in a pregnant patient who has a history of penicillin allergy?

- A) Azithromycin (Zithromax)
- B) Ceftriaxone
- C) Doxycycline
- D) Penicillin G benzathine (Bicillin L-A) after desensitization

158. A 7-year-old male with no significant past medical history presents with his parents for a well child visit. His BMI has consistently been above the 80th percentile. His parents have both struggled with maintaining healthy weights.

To help them promote a healthy lifestyle for their child, you should tell the parents which one of the following?

- A) An hour per day of moderate or vigorous physical activity is recommended starting at age 6
- B) Calorie restriction alone is sufficient in maintaining a healthy weight
- C) Children should avoid video screen-based physical activity
- D) Organic fruit juices can be provided to achieve the recommended daily servings of fruits and vegetables

159. A 56-year-old female presents with numbness of her bilateral feet. She first became aware of this during a recent pedicure when she had difficulty feeling the nail technician work on her toes and the plantar surface of her feet. Since that time, she has noted a feeling of "walking on cotton gauze" and has started wearing slippers around her home. She had intense pruritus of her feet approximately 6 months ago that persisted for several weeks, but it was not accompanied by paresthesia. She is otherwise in good health and participates in an aerobics class 3 days per week. She takes lisinopril (Zestril), 10 mg daily, for essential hypertension. She does not have any other symptoms.

On examination, her cranial nerves, muscle strength, and reflexes are normal. Sensation is diminished over the plantar surface and toes of both feet. She has 2+ dorsalis pedis and posterior tibial pulses bilaterally. Routine laboratory studies, including a CBC, comprehensive metabolic panel, hemoglobin A_{1c}, serum protein electrophoresis, and TSH, vitamin B₁₂, and folate levels, are all in normal range. She is concerned that she may be developing multiple sclerosis.

Which one of the following would be the most appropriate next step?

- A) Gabapentin (Neurontin), 100 mg 3 times daily
 - B) Vitamin B₆ (pyridoxine), 50 mg daily
 - C) Lyme antibody testing
 - D) Nerve conduction testing and electromyography (EMG)
 - E) Brain MRI
160. A 56-year-old female presents for evaluation of right heel pain. She recently returned from a hiking trip during which she developed progressive heel discomfort. The patient's only medication is losartan (Cozaar), which she takes for hypertension. After a thorough examination, you diagnose Achilles tendinopathy.

Which one of the following would be the most appropriate next step in diagnosis and management?

- A) MRI for confirmation of diagnosis
- B) Non-weight-bearing until pain is improved
- C) Corticosteroid injection of the Achilles tendon
- D) Platelet-rich plasma injection
- E) Referral for physical therapy

161. A 45-year-old male with a new diagnosis of type 2 diabetes is anxious about managing his condition and uncertain about his ability to meet the demands of treatment and self-care. Which one of the following would be the most appropriate strategy to help him understand his diagnosis and follow through on lifestyle and/or medication changes?
- A) Emphasizing the seriousness of the condition by showing pictures of patients with complications
 - B) Explaining the diagnosis and treatment plan with medical language to ensure professional communication
 - C) Supplying only detailed written instructions about his condition immediately after the appointment
 - D) Using motivational interviewing to identify what is most important to the patient and create impetus for change
162. A generally healthy 15-year-old female presents to your office with a history of frequent, generalized abdominal pain and loose stools daily for the last few months. She has not noted any relationship to food or a specific frequency. She has not had any previous surgeries and is up to date on routine vaccinations. At her 14-year-old well child visit, her height was at the 50th percentile and her weight was at the 60th percentile. Today, her height is at the 35th percentile and her weight at the 20th percentile. She has had a weight loss of 1 kg (2 lb). Her vital signs are normal.

Which one of the following would be the most appropriate next step?

- A) Reassurance and dietary monitoring to evaluate for a relationship to symptoms
- B) Fecal fat testing
- C) An IgA tissue transglutaminase level
- D) Carbohydrate breath testing
- E) A referral for esophagogastroduodenoscopy

163. A 60-year-old male asks you about testosterone supplementation during a routine office visit. He reports that he has had decreased energy, decreased libido, and loss of upper and lower body strength over the last year. He would like to have his testosterone level checked today.

Which one of the following symptoms, if present in this patient, would be most concerning for testosterone deficiency?

- A) Difficulties with memory
- B) Fatigue or diminished physical stamina
- C) Loss of axillary and pubic hair
- D) Thinning of hair at the temples
- E) Thinning or loss of eyebrows

164. A 3-year-old female is brought to your office by her parents for a preschool examination. She had a normal birth and is up to date on her vaccinations. On examination she appears to be healthy, and her development is appropriate for her age. She is in the 50th percentile for height and weight. You note a grade 1–2/6 continuous murmur best heard in the infraclavicular region when she is upright. The murmur changes slightly in intensity with head movement and is absent when she is supine.

Which one of the following would be the most appropriate next step?

- A) Reassuring the parents that this is a benign venous hum that needs no follow-up
- B) Scheduling a follow-up appointment in 1 month to recheck the murmur
- C) Recommending that the patient undergo echocardiography when she reaches adulthood
- D) Sending the patient for evaluation for repair of a patent ductus arteriosus
- E) Referral to a pediatric cardiologist

165. A 40-year-old female presents to your office after an episode of anaphylaxis. She has a known tree nut allergy and developed oral swelling, dyspnea, and chest pain after eating a muffin from a coffee shop. Her symptoms initially improved after she used her epinephrine auto-injector but began to worsen 10 minutes later. She used a second dose of her auto-injector 10 minutes prior to presentation and is feeling much better.

Which one of the following would be the most appropriate next step?

- A) Starting famotidine (Pepcid), 20 mg daily, and follow-up in 1 week
 - B) Starting prednisone, 40 mg for 5 days, and follow-up in 5 days
 - C) Starting fexofenadine (Allegra), 180 mg daily, and immediate follow-up with an allergist
 - D) Transfer to the emergency department (ED) with observation for 1–2 hours prior to discharge
 - E) Transfer to the ED with observation for 6–12 hours prior to discharge
166. A 42-year-old male has been admitted to the hospital for hypotension and electrolyte disturbances due to chronic diarrhea. He was stabilized in the ICU and transferred to the inpatient floor. His gastroenterologist is asking for a colonoscopy with biopsies. His medical history is significant for congenital heart disease with a mechanical aortic valve replacement, an embolic stroke resulting in right hemiparesis, a deep vein thrombosis, and a solitary kidney. He is currently anticoagulated with warfarin and has an INR of 2.6. He has a creatinine level of 2.8 mg/dL (N 0.7–1.3) and a glomerular filtration rate of 28 mL/min/1.73 m².

In addition to holding warfarin for the planned colonoscopy, how should you manage this patient's anticoagulation while he is hospitalized?

- A) Resuming warfarin post procedure with a goal INR of 2.0–3.0
- B) Vitamin K for reversal, and resuming warfarin post procedure with a goal INR of 2.5–3.5
- C) Enoxaparin (Lovenox), 1.5 mg/kg daily
- D) Heparin, 5000 units subcutaneously 3 times daily
- E) A heparin drip titrated to achieve therapeutic aPTT or anti-factor Xa levels

167. A 4-year-old male is brought to the emergency department by his parents after tripping and falling on his hand in an outstretched position. An x-ray reveals a buckle (torus) fracture of the distal radial metaphysis.

Which one of the following management strategies would be most appropriate?

- A) A short arm splint for 3 weeks with follow-up radiography
 - B) A short arm splint for 3 weeks with follow-up radiography only if pain or tenderness persists
 - C) A short arm cast for 6 weeks with follow-up radiography
 - D) A short arm cast for 6 weeks with follow-up radiography only if pain or tenderness persists
168. A 19-year-old female presents for a wellness visit while home on college break. According to the U.S. Preventive Services Task Force, which one of the following screenings is indicated?
- A) Cavities or periodontal disease
 - B) Dyslipidemia
 - C) Skin cancer
 - D) Suicide risk
 - E) Unhealthy drug use
169. A 33-year-old oyster fisherman presents because of a 3-day history of diarrhea. He describes frequent, large-volume, watery stool that is milky white. His vital signs are normal and he appears nontoxic.

Infection with which one of the following pathogens is most likely?

- A) *Clostridioides difficile*
- B) Enteropathogenic *Escherichia coli*
- C) Enterotoxigenic *E coli*
- D) *Vibrio* species
- E) *Yersinia enterocolitica*

170. A 35-year-old female with Down syndrome presents for a health maintenance examination. She does not have any new symptoms or concerns today. Her only medication is an etonogestrel implant (Nexplanon) that was placed 2 years ago. She is monogamous with a male sex partner. On examination you note normal vital signs with a BMI of 28 kg/m². At her most recent laboratory evaluation 2 years ago, she had an LDL-cholesterol level of 104 mg/dL, a fasting glucose level of 89 mg/dL, and a TSH level of 2.67 μ U/mL (N 0.4–4.2).

Which one of the following screening evaluations should you order today?

- A) The National Task Group–Early Detection Screen for Dementia tool
 - B) A fasting lipid panel
 - C) TSH testing
 - D) A DXA scan
 - E) Cervical spine radiography
171. You are participating in a continuing medical education (CME) course at a Colorado ski resort that is 9600 ft (2926 m) above sea level. You and your travel companions have had intermittent issues with headaches, loss of appetite, and difficulty sleeping. On the third day, one of your companions starts to experience shortness of breath that is not relieved with rest. Their symptoms worsen throughout the day and they become slightly tachycardic. Later that evening, they develop a cough that produces frothy sputum.

Which one of the following would be the most appropriate next step?

- A) Acetazolamide
 - B) Dexamethasone
 - C) Nifedipine
 - D) A COVID-19 test
 - E) Oxygen supplementation, and descending to a lower elevation
172. Which one of the following antibiotics is most commonly associated with Stevens-Johnson syndrome?
- A) Cephalexin
 - B) Ciprofloxacin
 - C) Fosfomycin
 - D) Nitrofurantoin
 - E) Sulfamethoxazole/trimethoprim

173. During an annual visit, a 62-year-old female reports an episode of vaginal spotting that lasted for a few days in May. It is now July and she has not had another recurrence. A transvaginal ultrasound shows an endometrial stripe of 4 mm, no fibroids, and no polyps.

Which one of the following would be most appropriate regarding follow-up evaluation?

- A) No further evaluation
- B) Endometrial biopsy
- C) Hysteroscopy and biopsy
- D) Dilation and curettage
- E) Repeat ultrasonography in 1 year

174. A 25-year-old male presents with shoulder pain after having fallen on his arm in an outstretched position. He reports a sensation of catching and popping in the shoulder. With the patient seated and hands placed on his hips, you apply force to the elbow and direct the humerus anteriorly and superiorly, while asking the patient to resist your motion, which produces pain in the affected shoulder.

Which one of the following is the most likely diagnosis?

- A) Acromioclavicular joint arthritis
- B) Adhesive capsulitis
- C) Glenohumeral osteoarthritis
- D) A glenoid labral tear
- E) A rotator cuff tear

175. According to the U.S. Preventive Services Task Force, which one of the following is the recommended next step for adults who screen positive for unhealthy alcohol use?

- A) Brief counseling interventions
- B) Cognitive behavioral therapy
- C) Mindfulness-based therapy
- D) Interpersonal psychotherapy
- E) Referral for alcohol use disorder treatment

176. A 30-year-old female presents with constipation and severe anal pain following bowel movements. On examination, you observe an anal fissure.

In addition to increasing fluids, fiber, and warm sitz baths, which one of the following would be effective and least likely to result in side effects?

- A) Topical calcium channel blockers
- B) Topical capsaicin
- C) Topical nitroglycerin
- D) Oral calcium channel blockers

177. Which one of the following interventions lowers the risk of diverticulitis?

- A) Avoiding consumption of popcorn, nuts, and seeds
- B) Regular consumption of red meat
- C) A vegetarian diet
- D) Long-term NSAID use

178. An 82-year-old male presents to your office because he has been experiencing severe, stabbing pain that feels like an electric shock on the left side of his face in his cheek, lips, and chin. The episodes are usually triggered by light touch or eating and last for a few seconds.

Which one of the following cranial nerves is associated with this disorder?

- A) V
- B) VII
- C) X
- D) XI
- E) XII

179. Which one of the following social determinants of health has a protective effective against Alzheimer disease and related types of dementia?

- A) Decreasing BMI with older age
- B) Early-life adversity
- C) Low education level
- D) Social engagement

180. A 72-year-old male with a history of type 2 diabetes comes to your office for evaluation of muscle aches and fatigue. His current medications include rosuvastatin (Crestor), dapagliflozin (Farxiga), and semaglutide (Ozempic).

Over the past month he has noticed increasing pain in his hips, thighs, and shoulders without any apparent trigger. He feels so stiff in the morning that it is becoming difficult to get out of bed. On examination his strength is diminished (4/5) in the proximal muscle groups. You do not note any swelling or redness of any joints, muscles, or extremities. He has normal reflexes and no clonus.

Laboratory Findings

Hemoglobin A _{1c}	6.4%
Hemoglobin	12.2 g/dL (N 13.2–16.6)
Creatinine	1.1 mg/dL (N 0.7–1.3)
Potassium	4.2 mEq/L (N 3.5–5.1)
Creatine kinase	84 U/L (N 55–170)
Erythrocyte sedimentation rate	88 mm/hr (N 0–20)
C-reactive protein	9.2 mg/L (N 0–0.3)
Antinuclear antibody	<1:40 (N <1:40)

Which one of the following is the most likely cause of this patient's myalgias?

- A) Fibromyalgia
- B) Polymyalgia rheumatica
- C) Polymyositis
- D) Statin-induced myopathy
- E) Systemic lupus erythematosus

181. The U.S. Preventive Services Task Force recommends which one of the following regarding screening asymptomatic adolescents and adults for herpes simplex virus?

- A) No testing
- B) HSV-1 antibody testing
- C) HSV-2 antibody testing
- D) HSV-1 and HSV-2 antibody testing
- E) HSV-1 and HSV-2 antigen testing

182. A 2-year-old child is brought to the urgent care clinic by their parents with a 2-day history of abdominal pain and vomiting. The child has been irritable over the past 2 days with intermittent abdominal pain and several episodes of associated emesis, and just had a bloody stool that the parents describe as resembling jelly. The child appears uncomfortable but is alert and afebrile, with normal blood pressure.

Which one of the following is the most likely diagnosis?

- A) Intussusception
- B) Necrotizing enterocolitis
- C) Pyloric stenosis
- D) Volvulus

183. A generally healthy 54-year-old male with a 20-pack-year smoking history presents to discuss smoking cessation. His first grandchild was just born and he is eager for help in quitting cigarette smoking. He quit smoking 2 years ago using nicotine gum (Nicorette Gum), but resumed smoking 5 months later. He has also explored telephone helplines and support groups, but did not find them beneficial. He currently smokes 1 pack of cigarettes per day.

Which one of the following would be most likely to help this patient quit smoking?

- A) Nicotine gum only
- B) Extended-release bupropion (Wellbutrin SR) only
- C) Varenicline only
- D) Nicotine gum plus extended-release bupropion
- E) Nicotine gum plus varenicline

184. A 60-year-old female returns to discuss management of her recurrent nephrolithiasis. She has a history of calcium stones and just recently passed her fourth stone this year despite appropriate lifestyle changes, including a fluid intake of 3 L per day and a diet high in fiber and low in animal protein. She would like a medication to help prevent the development of more kidney stones. She has a past medical history of asthma for which she takes a budesonide/formoterol inhaler (Symbicort) as needed. Today, her vital signs and a physical examination are normal. A recent basic metabolic panel was normal.

Which one of the following would be the most appropriate prevention strategy?

- A) Lifestyle modifications only
- B) Furosemide (Lasix), 20 mg daily
- C) Hydrochlorothiazide, 50 mg daily
- D) Magnesium oxide, 400 mg twice daily
- E) Tamsulosin (Flomax), 0.4 mg daily

185. A 72-year-old female is brought to the emergency department after her neighbor found her on the floor in her home. She is drowsy and confused, and unable to provide any history. An initial laboratory evaluation reveals the following:

Glucose	95 mg/dL
BUN	9 mg/dL (N 6–20)
Creatinine	0.9 mg/dL (N 0.5–1.1)
Estimated glomerular filtration rate	63 mL/min/1.73 m ²
Sodium	118 mEq/L (N 135–145)
Potassium	3.0 mEq/L (N 3.3–5.0)
Chloride	88 mEq/L (N 101–115)
Bicarbonate	22 mEq/L (N 16–30)
Anion gap	8 mEq/L (N 5–14)
Calcium	8.8 mg/dL (N 8.4–10.5)

Which one of the following would be the most appropriate intervention at this time?

- A) Free water restriction
- B) Normal saline infusion
- C) 3% hypertonic saline bolus
- D) Intravenous desmopressin (DDAVP)

186. A 35-year-old construction worker presents to your office with an acute onset of lower back pain that began 3 days ago. The patient describes the pain as a burning sensation that radiates down his left leg. He has no bowel or bladder dysfunction and has been in good health. A straight leg-raising test is positive at 45° on the left without motor deficit.

Which one of the following does evidence support as the first step in treatment of this patient's condition?

- A) Bed rest for 3 days
- B) Mobilization and manipulation
- C) A short-term corticosteroid
- D) Cyclobenzaprine
- E) MRI

187. A 46-year-old male with chronic hypertension presents to your clinic for routine follow-up. Due to issues with insurance coverage, he has not been taking his antihypertensive medications for the last 2 months. Today, he is asymptomatic and an examination is normal, except for a blood pressure of 162/100 mm Hg, which is confirmed on repeat testing.

In addition to addressing insurance and social determinants of health, which one of the following would be most appropriate for this patient?

- A) Monotherapy with a calcium channel blocker (CCB)
- B) Monotherapy with a thiazide-like diuretic
- C) Combination therapy with a CCB and an ACE inhibitor
- D) Combination therapy with a diuretic and a β -blocker
- E) Combination therapy with a diuretic and an aldosterone antagonist

188. A 14-year-old female presents to your office with her mother who is concerned about her daughter's recent 15-lb weight loss and severe fatigue. After ruling out other etiologies, you diagnose anorexia nervosa.

Which one of the following findings, if present in this patient, would indicate the need for inpatient care?

- A) Hair loss
- B) Muscle atrophy
- C) Parotid gland enlargement
- D) A potassium level of 2.6 mEq/L (N 3.5–4.9)
- E) A sodium level of 146 mEq/L (N 134–145)

189. A 30-year-old female presents to your office for an annual wellness visit. She reports a family history of Lynch syndrome and recently tested positive herself.

Which one of the following should you recommend regarding colon cancer screening in this patient?

- A) Fecal occult blood testing annually
- B) Stool DNA testing (Cologuard) every 3 years
- C) Colonoscopy starting this year
- D) Colonoscopy starting at age 45
- E) Sigmoidoscopy

190. A 53-year-old male without a significant past medical history presents to the emergency department for evaluation of abdominal pain, nausea, and vomiting. Pertinent findings on presentation include notable tenderness in the epigastrium and a lipase level >3 times the upper limit of normal. He is admitted for acute pancreatitis. Intravenous lactated Ringer solution, antiemetics, and pain control are initiated. At the time of your evaluation he has normal vital signs and reports improved symptoms, although you still note abdominal tenderness.

When entering admission orders for this patient, which one of the following diets would be most appropriate?

- A) Full liquids
- B) Low-carbohydrate
- C) Low-fat
- D) NPO

191. A healthy 57-year-old female presents for an annual health maintenance visit. Her last menstrual period was 2 years ago. She exercises 5 times per week, does not smoke, and rarely drinks alcohol. Her BMI is 23 kg/m². Her 80-year-old mother has no history of fractures, but takes alendronate (Fosamax), 70 mg weekly.

According to the U.S. Preventive Services Task Force, which one of the following would be the preferred strategy to screen for osteoporosis in this patient?

- A) Using the FRAX tool now to determine the need for DXA screening
 - B) Using the FRAX tool at age 65 to determine the need for DXA screening
 - C) Screening now with DXA
 - D) Screening at age 65 with DXA
192. You are evaluating a patient with acute-on-chronic kidney disease and a serum potassium level of 6.5 mEq/L (N 3.3–5.0). In addition to peaked T waves, which one of the following additional electrocardiogram changes is most likely to be present?
- A) Delta waves
 - B) Prolonged QRS interval
 - C) Prolonged QT interval
 - D) ST segment depression
 - E) U waves
193. A 15-year-old male comes to your clinic 30 minutes after being hit in the face by a baseball during practice. He is alert and answers questions appropriately. After clearing the blood, you find that his upper front tooth (No. 8) is quite loose and the root is visible. The tooth is lying lower than his other teeth. The tooth appears intact and without fracture. There is no crepitus or pain when palpating the alveolar and mandibular bones. You immediately call his dentist to arrange an emergency appointment.
- Which one of the following would be the most appropriate next step prior to sending the patient to the dentist's office?
- A) No intervention is needed
 - B) Repositioning the tooth in the socket and having the patient bite down on gauze
 - C) Gently removing the tooth and placing it in saline or milk
 - D) Ceftriaxone, 1 g intramuscularly
 - E) Imaging of the facial bones

194. A 50-year-old female presents for follow-up of laboratory studies. At her last visit, she presented with palpitations, anxiety, and unintentional weight loss. An examination was notable for exophthalmos. Her laboratory results show a very low TSH level with elevated free T₄ and normal total T₃ levels. A radioactive iodine uptake scan demonstrates a high, homogeneous uptake of iodine.

Which one of the following medications is the most appropriate treatment for this condition?

- A) Levothyroxine (Synthroid)
- B) Methimazole
- C) Prednisone
- D) Propylthiouracil
- E) Verapamil

195. An 85-year-old male on the third day of hospitalization for the treatment of pneumonia develops symptoms of altered mental status. Evaluation of his medications does not identify a reason for his altered mental status. The nursing staff has been unable to successfully redirect him to keep him calm.

Which one of the following medications would be appropriate for the management of the patient's delirium?

- A) Haloperidol
- B) Lithium
- C) Lorazepam (Ativan)
- D) Olanzapine (Zyprexa)
- E) Oxycodone

196. An 84-year-old male with a history of coronary artery disease, mild cognitive impairment, and type 2 diabetes is admitted to your hospital for a hip fracture after a recent fall. On examination the patient has normal speech but is confused. He is clearly unable to articulate his understanding of the risks and benefits of surgery.

You should do which one of the following to determine this patient's decision-making capacity regarding surgery?

- A) Administer the Aid to Capacity Evaluation assessment tool
 - B) Administer the Montreal Cognitive Assessment evaluation tool
 - C) Document a lack of capacity and consult his designated durable power of attorney
 - D) Consult with psychiatry
 - E) Defer to the court
197. A 36-year-old backcountry skier becomes separated from her group during a blizzard. The ski patrol locates her several hours later, sitting on the ground and not moving or shivering. She is conscious but confused.

Which one of the following would be the most important initial step?

- A) Obtaining a core body temperature
 - B) Providing a warm drink and high-calorie snack, and reconnecting her with her group
 - C) Having her remove her wet clothes and wrapping her in an insulated vapor barrier
 - D) Placing large heat pads around her upper body and wrapping her in an insulated vapor barrier
 - E) Performing emergency rapid sequence field intubation and preparing for evacuation
198. A 63-year-old male presents for his annual health maintenance examination. He maintains a healthy lifestyle, eats well, and exercises regularly. A physical examination is normal. His most recent prostate cancer screening revealed a PSA level of 4.6 ng/mL (N 0.3–4.0), and his previous result 2 years ago was normal.

Which one of the following would be the most appropriate next step in management?

- A) Repeating PSA testing in 4–8 weeks
- B) Empiric antibiotic therapy
- C) A digital rectal examination
- D) Biopsy of the prostate
- E) MRI of the prostate

199. An otherwise healthy 37-year-old female presents because of a 4-day history of multiple episodes of watery, nonbloody diarrhea. She has no fever, chills, abdominal pain, or vomiting. She does not have a history of recent antibiotic use, hospitalization, or travel, but does say that her teenage son recently returned from a spring break trip and has similar symptoms. Physical examination findings include dry mucous membranes and tachycardia. The abdomen is soft with mild generalized tenderness on deep palpation.

Which one of the following next steps would be most appropriate at this time?

- A) No testing
- B) Fecal calprotectin
- C) Fecal lactoferrin
- D) Fecal leukocytes

200. Which one of the following dog bite wounds should be considered for primary closure instead of being left to heal by secondary intention?

- A) A crush injury to the arm
- B) A gaping wound to the hand
- C) A linear wound to the face
- D) A puncture wound to the leg